Ichthyop User Guide (3.5)

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Preface

Part I

User guide

1 Getting started

In this section, download and install instructions are provided.

1.1 Prerequisites

1.1.1 Java

In order to run Ichthyop, **Java (>= 11)** needs to be installed. To make sure that Java is installed and that you have the right version, type:

```
java --version
```

If you see an error, then you need to install Java. If the version is less than 11, you need to reinstall it.

If needed, see Chapter 8 for the Java install instructions.

1.1.2 NetCDF4

The Java library that manages input/output of files in Ichthyop requires the external NetCDF C library, which can be installed as follows:

1.1.2.1 Mac Os X

Open a Terminal and type:

```
sudo port install netcdf4
```

1.1.2.2 Linux

Open a Terminal and type:

```
sudo apt-get install netcdf4
```

1.1.2.3 Windows

Download the pre-built libraries from Unidata website

> Caution
>
> During the install process, make sure that the location of the library is added to the PATH

1.1.3 Conda environmenmt

There is also the possibility to use Conda environments in order to install Maven, OpenJDK and NetCDF4 easily. Instructions can be found on https://github.com/ichthyop/ichthyop-conda

1.2 Downloading Ichthyop

The Ichthyop tool is available on GitHub. There are two ways to use Ichthyop:

- Use executable files (`.jar` files) (**preferably for beginners**)
- Use source files (**preferably for advanced users and developers**)

1.2.1 Using executable files

Ichthyop executable files can be downloaded here. Download the latest `ichthyop-X.Y.Z-jar-with-dependencies.jar` file (replacing `X.Y.Z` by the version number). Older versions are also available if necessary.

1.2.2 From source

To get the source code, type in a Terminal (Unix/MacOs) or Git Bash prompt (Windows):

```
git clone https://github.com/ichthyop/ichthyop.git
```

The code can then be compiled either using IDE (NetBeans, VSCode) or using the following command line:

```
mvn package
```

The executable will be generated in the `target` folder.

> **Warning**
>
> To use the command line, Maven needs to be installed (see instructions on https://maven.apache.org/install.html)

1.3 Downloading input hydrodynamic files

To run Ichthyop hydrodynamic files with (at least) water velocity fields are required. Demo files generated by several hydrodynamic models can be downloaded here. Put these files into the ichthyop/input folder.

1.4 Running Ichthyop

1.4.1 Clicking on file (Windows)

Open the Ichthyop folder and double click on the `ichthyop-X.Y.Z-jar-with-dependencies.jar` file, where `X.Y.Z` is the Ichthyop version. The Ichthyop console should open up

1.4.2 From command line (Unix/Mac Os X)

Open a command line prompt (Terminal or CMD prompt) and navigate to the Ichthyop folder using `cd`.

Then, type:

```
java -jar ichthyop-X.Y.Z-jar-with-dependencies.jar
```

with `X.Y.Z` the Ichthyop version.

This will prompt the Java console. In order to run Ichthyop without the console, you need to specify a supplementary argument, which is the XML configuration file (see Chapter 3)

```
java -jar ichthyop-X.Y.Z-jar-with-dependencies.jar cfg-roms3d.xml
```

2 Ichthyop user interface (beginners)

When Ichthyop is ran with its user interface, a simulation comprises 4 steps

- configuration (Section 2.1)
- running (Section 2.2)
- visualize results (Section 2.3)
- and animation (Section 2.4).

2.1 Configuration

Firstly you will find the usual “File menu” functions : create a new configuration file, open an existing one, close it and save it.

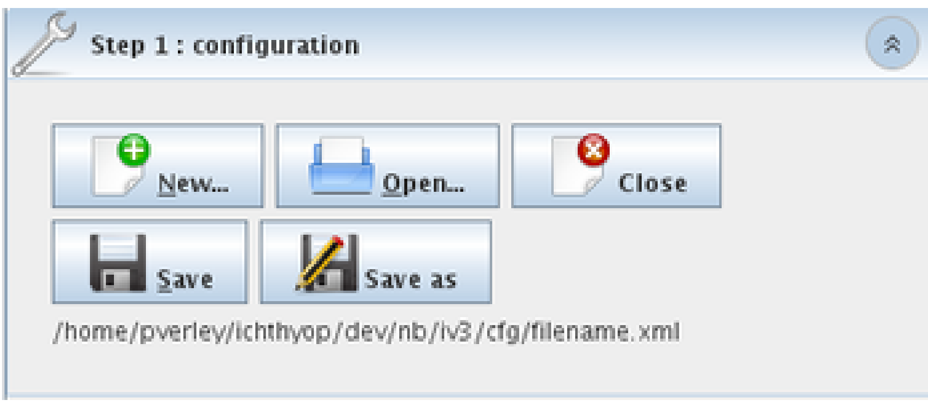


Figure 2.1: Step 1, configure

2.1.1 New configuration file

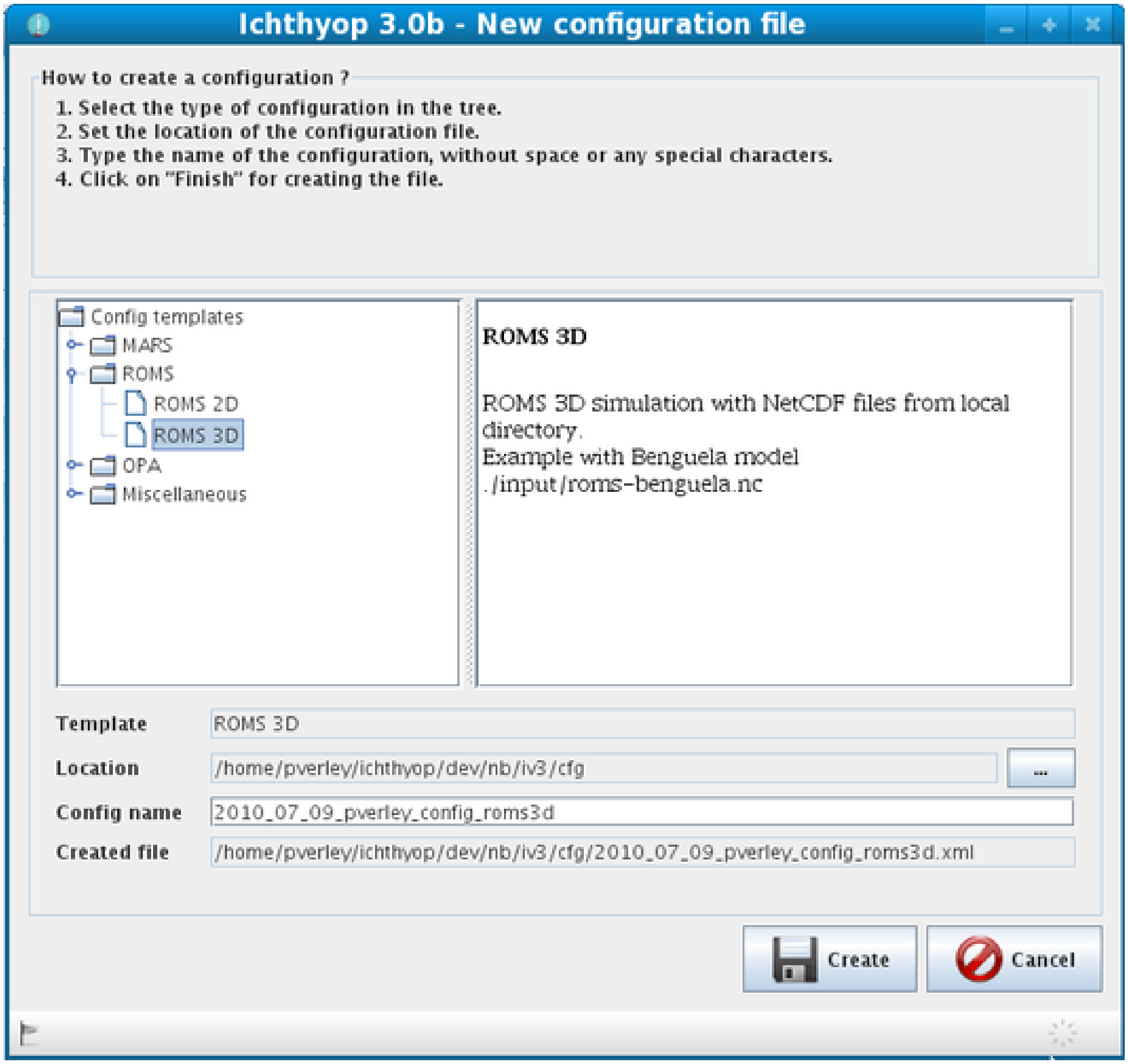


Figure 2.2: Step 1, create a new configuration file

The application comes with some preset examples of configuration files (i.e. the templates). Select one of the templates, change the name of the configuration if the suggested name does not suit you and click on the Create button.

2.1.2 Content of the configuration file

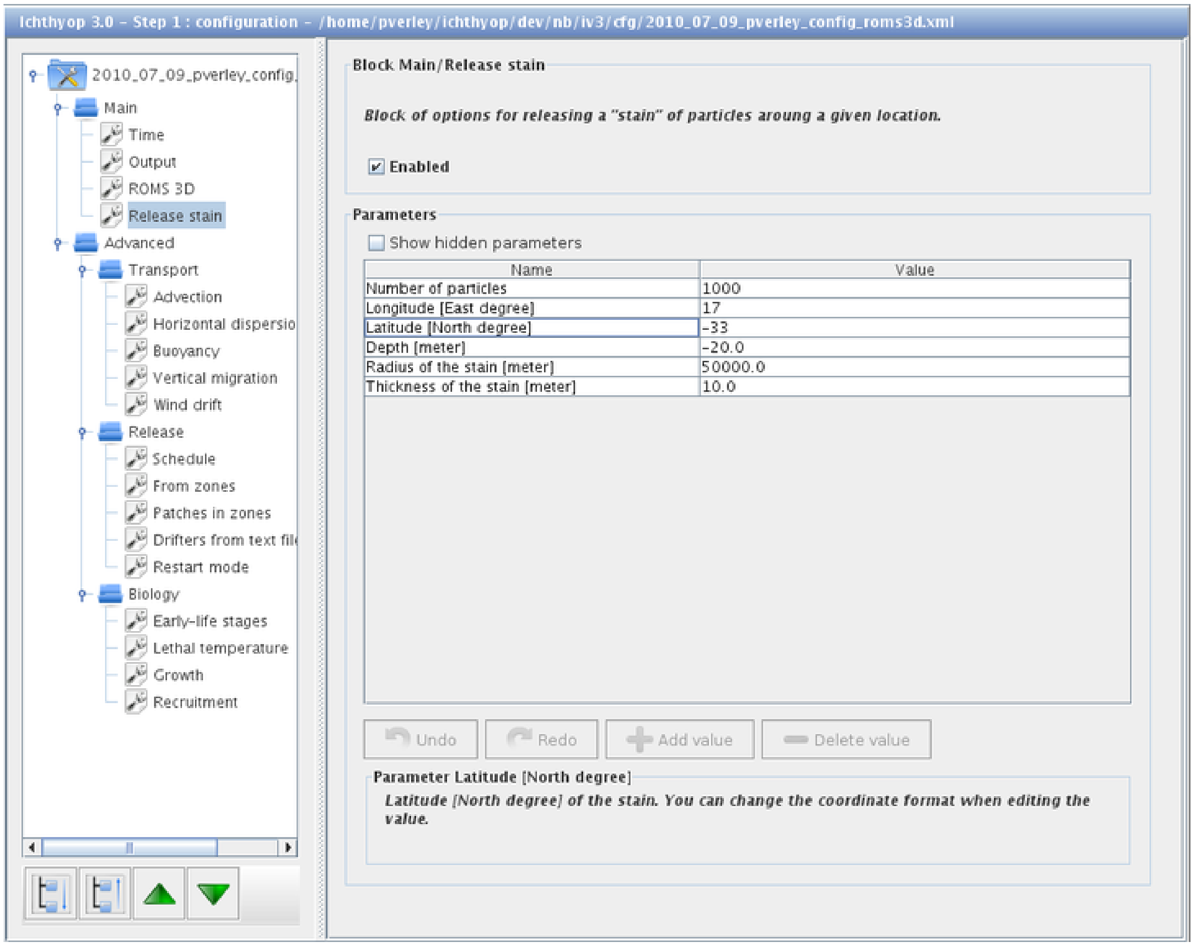


Figure 2.3: Step 1, edit the configuration file

The configuration file is organized in several categories and each category contains several blocks of parameters.

Main blocks:

- Time: Set the simulation time options, such as the beginning of the simulation, the duration of transport, the time step, etc.
- Output: Options that control the record of the particle tracks in a NetCDF output file.
- Dataset (e.g. ROMS 3D): Management of the hydrodynamic dataset (which comes as inputs of Ichthyop).
- Release: Determine how and where the particles should be released (see Chapter 5)

Advanced blocks:

- Transport: Parameters for controlling the advection process, the dispersion, the vertical migration, the wind drift, etc.
- Release: Additional ways for releasing particles, from zones, with position recorded in a text file or in a NetDCF file, etc.
- Biology: Control the biological processes such as growth or mortality.

Each block is described in the block information area. You may also find a description of parameters in the parameter information area.

> Warning
>
> Always save the configuration file before going to the next step.

> Caution
>
> **Only paramters that are already defined in a comfiguration file can be edited using the console**. Advanced users wishing to add new parameters are referred to Chapter 3.

> Caution
>
> **Using the user interface, additionnal values can be provided to serial parameters only when hidden parameters are displayed**

2.1.3 Zone definition

Users can define geographical zones either for "release" (i.e., origin) or "recruitment" (i.e., destination) of particles.

The first time the user enable (`Enabled` tick box) zones for release (into block `Release - From zones`) or for recruitment (into block `Biology - Recruitment - In zones`), do not forget to create a new zone file (by clicking `New` in the zone file value field) as the filename is preset in the configuration editor but the file itself may not exist yet on your disk. After that the user will be able to edit an existing zone file by clicking `Edit` in the zone file value field.

Zones can be added, edited or removed directly within the user interface (Figure 2.4).

2.1.3.1 Adding, removing and renaming zones

The number of zones is managed on the left part of the panel.

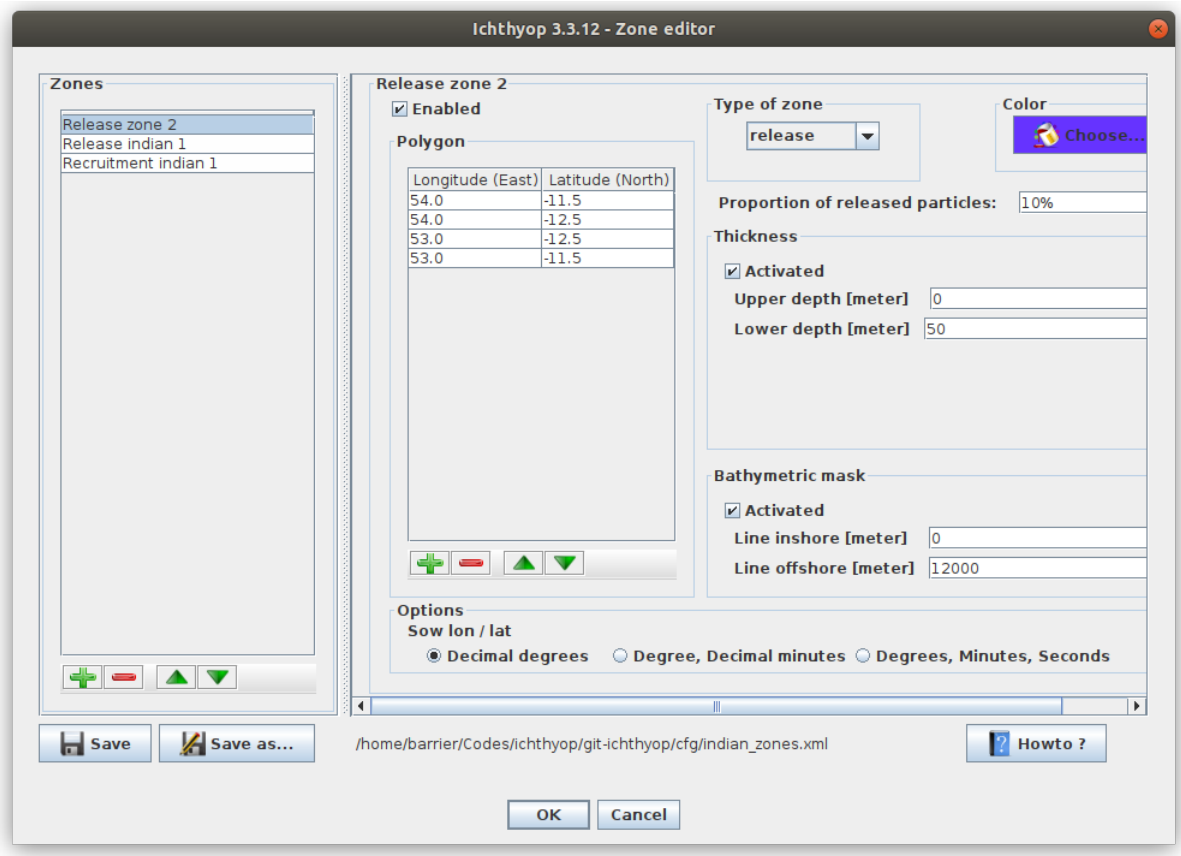
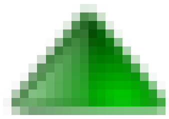
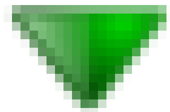


Figure 2.4: Ichthyop Zone editor

New zones can be added by clicking the  button. When a zone is selected, it can be removed by clicking on the button.

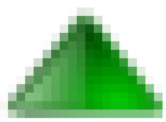
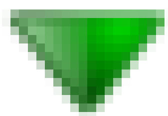
The reordering of the zones is achieved by clicking on the  and  buttons.

When double-clicking on the name of the zone on the left panel, the user rename a zone.

2.1.3.2 Editing a zone

When a zone is selected, the user can edit different parameters associated with the zone.

First, the user can enable or disable a zone by clicking on the `Enabled` tick box.

The zones are defined by providing the geographic points coordinates. Points can be added, removed and reordered by using the , ,  and  buttons, respectively. The user can also change the format of the points coordinates by clicking on the radio buttons in the `Options` bottom panel.

Ichthyop uses three types of zones: one for release (see Section 5.2), one for recruitment of particles and one for orientation (see Section 7.3.1). This type is chosen by using the `Type of zone` combo box. For release zones, the user can specify the number of particles that will be released in the zone (only if the `user_defined_nparticles` parameter is set equal to true, cf Section 5.2). This is done by filling the `Proportion of released particles` textbox and pressing `ENTER`

Each zone is associated with a color for the preview and the display of the simulation results. This color can be edited by using the  button.

In the case of 3D simulations, the user can specify a subset of zone area based on a lower and upper depth. This feature is activated by clicking on the `Activated` tick box of the `Thickness` panel. The user can also specify a subset of zone area based on bathymetry (for example 0 to 200m, for using only the subset of zone area from 0 to 200 m depth). This feature is activated by clicking on the `Activated` tick box of the `Bathymetric mask` panel.

2.2 Running

Figure 2.5: Step 2, simulation

The user may first want to preview the simulated area by clicking on the "Preview" button. This is also a way to check that the simulation is correctly set up, i.e. that the application found all the parameters required for starting the simulation. More specifically, Ichthyop will attempt to read the geographical variables (longitude, latitude and depth) from the hydrodynamic dataset in order to draw the area. It should also display the release and the recruitment zones if they have been defined and activated in the configuration file. Make sure that what you see is what you expect, and go back to "Step 1: configure" if not.

When the preview is satisfactory, click on "Run simulation" for starting the simulation. The progress bar will give an estimation of the remaining time for the simulation to complete. You can interrupt (but not pause) the simulation anytime by a click on "Stop simulation".

Depending on the capabilities of your computer, the number of released particles, how many actions are enabled, etc. the simulation may require a large amount of the available dynamic memory and the application may look like it is frozen. Wait until the simulation runs to completion. Refer to Section 4.1 if the application crashes because of memory issue.

2.3 Visualize results

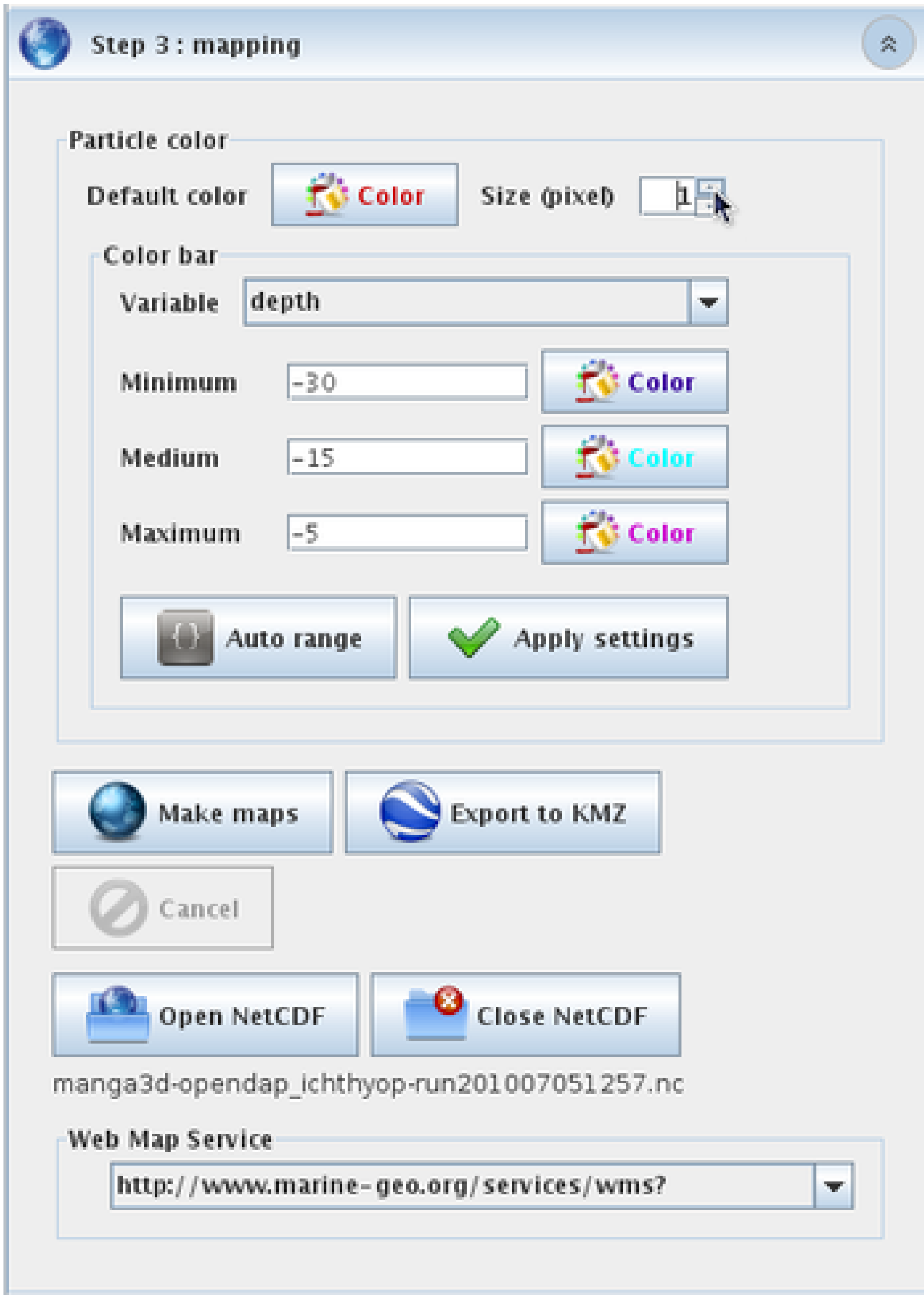


Figure 2.6: Step 3, mapping

When the simulation is completed, the application automatically opens the current Ichthyop output file for visualizing the results. If your computer is connected to Internet, you should see the map being centered above the simulated area. Otherwise, it only displays a Grey background.

Click on "Open NetCDF" and select the Ichthyop output file you wish to visualize. When the NetCDF file is opened, the application brings you back to the exact point where it was when the simulation just completed.

The application offers two ways for visualizing the results :

- `Make maps`, i.e. draw the successive locations of the particles on a map over time
- Export the particle trajectories in a KMZ file that can be opened with Google Earth.

2.3.1 Results

Ichthyop archives the simulation outputs in NetCDF format, a machine-independent data formats for sharing array-oriented scientific data.

The NetCDF file is recorded in the Ichthyop output folder (set in the `Output` section of the configuration file) and the file name contains the date and time of the file creation.

Default content of the NetCDF output file is the time and, for each particle, the longitude and latitude, the mortality status and in case of 3D simulations, the depth

2.3.2 Particle color

The `Default color` button allows to set the particle color for visualizing the trajectories.

Particles are plotted as small circles. `Particle size` determines the diameter of the circle in pixel.

To use a colorbar, select in the Combo box a variable archived in the Ichthyop output file (e.g. depth or time) you wish to visualize as a tri-color range. The `Auto range` button will scan the values of this variable and suggest the following range : `[mean - 2 * standard deviation; mean + 2 * standard deviation]`. Do not forget to click on `Apply settings` for validating the changes of the color bar.

For taking off the color bar, select the `None` item in the Combo box and click on `Apply settings`. A click on `Default color` button also deactivates the color bar.

2.3.3 Make maps using Web Map Service

A Web Map Service (WMS) is a standard protocol for serving georeferenced map images over the Internet.

Ichthyop provides two different WMS for displaying the ocean bathymetry and the cost line as a background of the particle trajectories.

Maps can be intuitively zoomed in and out with the mouse wheel and re-centered doing a mere drag and drop.

Depending on the quality of the Internet connection and how busy is the Web Map Server, the display of the background tiles may take a while or even not work at all. In that case, try again with another WMS and change the zoom scale.

When the settings of the map look satisfactory, click on the `Make maps` button. Ichthyop will then create a folder that has exactly the same name as the simulation NetCDF output file (without the .nc extension) in the output directory and save the maps in this folder as PNG images.

Again, depending on the computer capabilities and the number of particles, the map creation may require a large amount of the available dynamic memory and the application may seem frozen. Wait for the application to complete this step. Refer to Section 4.1 if the application crashes because of memory issue.

2.3.4 Export trajectories to KMZ format

Click on `Export to KMZ` button for recording the particle trajectories into a KMZ file. The file is recorded in the same directory as the Ichthyop output file, with the same name and the `.kmz` extension. KMZ format is the standard file format for visualizing georeferenced information with GoogleEarth.

Color settings (default color or color bar) and particle size will also be stored in the KMZ file.

When the export is completed, browse to the output folder and click on the KMZ file for launching GoogleEarth (assuming the program is already installed on your computer).

2.4 Animation

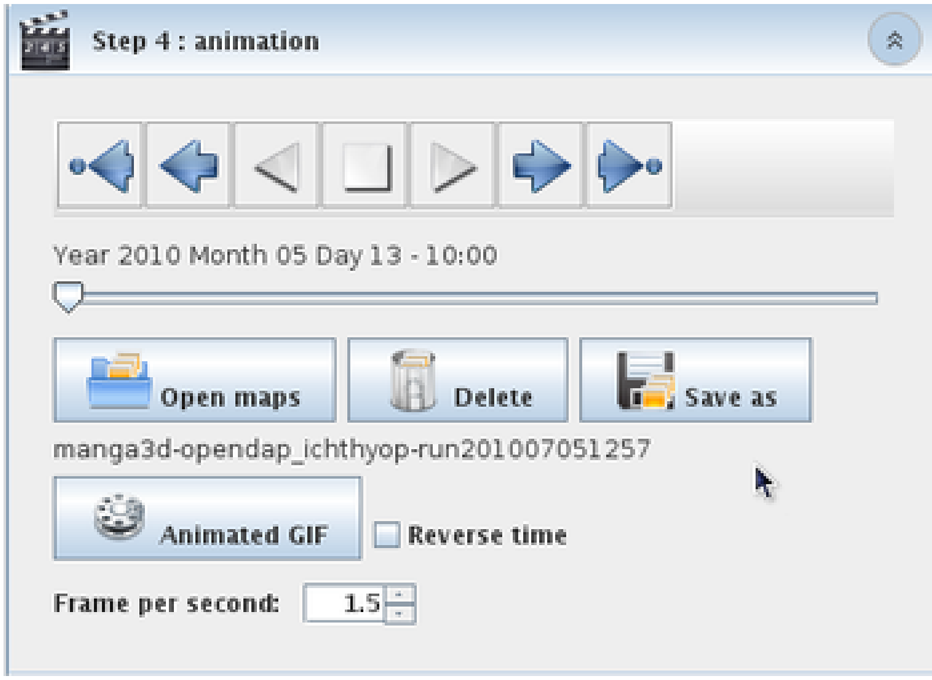


Figure 2.7: Step 4, Animation

Click on "Open maps" and select the simulation output folder that contains the PNG pictures you wish to visualize. When the folder is opened, the application brings you back to the exact point where it was when the map creation just completed.

Set the number of frames per second of the animation with the spinner.

You can also create an animated GIF. The file is recorded in the same directory as the Ichthyop output file, with the same name and the ".gif" extension.

3 Ichthyop configuration (advanced users)

Advanced users may want to change the configuration files directly rather than via the graphic interface.

3.1 Simulation configuration file

Ichthyop simulations are configured using XML configuration files. These files should always start as follows:

```
<icstructure>
<long_name>Generic Ichthyop configuration file</long_name>
<description>The file has few pre-defined parameters.</description>

</icstructure>
```

3.1.1 Configuration blocks

Ichthyop is configured by blocks, each block managing a specific aspect of the model. The blocks are as follows:

- ACTION: Block of parameters related to the `action` classes (cf. Chapter 7).
- RELEASE: Block of parameters related to the `release` classes (cf. Chapter 5)
- DATASET: Block of parameters related to the `datasets` classes.
- OPTION: Block of parameters related to the remaining parameters.

New blocks can be added in the XML file as follows:

```
<block type="option">
        <key>app.transport</key>
        <enabled>true</enabled>
        <tree_path>Transport/General</tree_path>
        <description>Set the general transport options of the simulation.</description>
</block>
```

The <key> tag is used to identify the configuration block. The <tree_path> tag is used in the Ichthyop console to create the parameter tree. The <description> field is used to display the block description in the Ichthyop console.

The <enabled> tag specifies whether the block should be considered when running the Ichthyop code or not. By default, all blocks are enabled. But for the RELEASE and DATASET blocks, one and only one block must be enabled.

3.1.2 Configuration parameters

A list of parameters is associated to each block. This is added in the XML file as follows:

```xml
<parameters>
</parameters>
```

Inside the <parameters> tag, new parameters are defined as follows:

```xml
<parameter>
    <key>output_path</key>
    <value>output</value>
    <long_name>Output path</long_name>
    <format>path</format>
    <default>output</default>
    <description>Select the folder where the simulation NetCDF output file should be saved.</desc
</parameter>
```

The <key> tag allows to identify the parameter, while the <value> tag specifies the value of this parameter. The <default> tag provides the default value of the parameter. The <long_name> and <description> tags are used by the console to provide informations about the parameter.

The <format> tag specifies the parameter format, which will be used by the console parameter editor. The accepted values are:

- `path`: For files and folders
- `date`: For dates (format must be `year YYYY month MM day at HH:MM`)
- `duration`: For duration (format must be `###### day(s) ### hour(s) ### minute(s)`)
- `float`: For real values
- `integer`: For integer values.
- `class`: For class parameters. It allow the user to choose an existing Ichthyop class in the configuration file.
- `list`: For a list of string parameters, separated by ,
- `boolean`: For boolean parameters. It allows the user to select `true` or `false` using a simple combo box.

- `combo`: For parameters with a limited set of values, which can be selected in the console with a combo box.
- `lonlat`: For geographical coordinates.

In the case of `combo` parameters, the list of accepted parameters is specified by providing as many `<accepted>` tags as necessary. For example:

```
<parameter>
    <key>time_arrow</key>
    <long_name>Direction of the simulation</long_name>
    <value>forward</value>
    <format>combo</format>
    <accepted>backward</accepted>
    <accepted>forward</accepted>
    <default>forward</default>
    <description>Run the simulation backward or forward in time.</description>
</parameter>
```

If a parameter should by default appears as hidden in the Ichthyop console, it can be specified by adding the `hidden="true"` argument to the `parameter` tag, as shown below:

```
<parameter hidden="true">
</parameter>
```

3.1.3 Serial parameters

In order to test different values of a given parameter, a `serial` tag can be added as follows:

```
<parameter type="serial">
</parameter>
```

Different values are then provided by replicating the `value` parameter as follows:

```
<parameter type="serial">
  <key>initial_time</key>
  <long_name>Beginning of simulation</long_name>
  <value>year 2001 month 10 day 20 at 00:00</value>
  <value>year 2001 month 10 day 21 at 00:00</value>
  <format>date</format>
  <description>Set the beginning date and time of the simulation. Format: year ###### month ### d
</parameter>
```

When simulations are run with serial parameters, all possible combinations of parameter values will be run. Output files will then contain a _sX suffix, with X the simulation number. The parameters used in a particular simulation X are provided as global NetCDF attributes in the corresponding _sX NetCDF output file.

3.2 Zone configuration file

Zone configuration files are also managed via a dedicated XML files. These files must be as follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<zones>

</zones>
```

Each zone is defined into a <zone> tag, which contain the following tags:

- <key> is the name of the zone
- <enabled> specifies whether this zone must be activated by default or not.
- <type> specifies whether the zone should be used for release (release value, see Section 5.2) or recruitment (recruitment value)
- <polygon> specifies the different points used to define the zond area
- <bathy_mask> (optional) specifies a subset of zone area based on bathymetry (for example 0 to 200m, for using only the subset of zone area from 0 to 200 m depth)
- <thickness> (optional) specifies the upper and lower depths subsets of zone areas (**only valid for 3D runs**).
- <color> specifies the display color of the zone (format is [r=102,g=51,b=255]).
- <proportion_particles> (optional) specifies the percentage of total particles (values in [0 – 1]) to be released in the area. This attribute is only used when type is release and if the user_defined_nparticles parameter is set to True (cf. Section 5.2)

> ⚠ Warning
>
> The sum of the values of proportion_particles over all the release zones should be 1!

An example of a zone definition is provided below.

```
<zone>
    <key>Release zone 2</key>
    <enabled>true</enabled>
    <type>release</type>
    <polygon>
```

```xml
        <point>
            <index>0</index>
            <lon>54.0</lon>
            <lat>-11.5</lat>
        </point>
        <point>
            <index>1</index>
            <lon>54.0</lon>
            <lat>-12.5</lat>
        </point>
        <point>
            <index>2</index>
            <lon>53.0</lon>
            <lat>-12.5</lat>
        </point>
        <point>
            <index>3</index>
            <lon>53.0</lon>
            <lat>-11.5</lat>
        </point>
    </polygon>
    <bathy_mask>
        <enabled>true</enabled>
        <line_inshore>0.0</line_inshore>
        <line_offshore>12000.0</line_offshore>
    </bathy_mask>
    <thickness>
        <enabled>true</enabled>
        <upper_depth>0.0</upper_depth>
        <lower_depth>50.0</lower_depth>
        </thickness>
    <color>[r=102,g=51,b=255]</color>
    <proportion_particles>0.2</proportion_particles>
</zone>
```

3.3 Time configuration

3.3.1 Beginning of the simulation

In Ichthyop, the user provides the time at which the simulation should start. This `initial_time` parameter, which must be defined in the `app.time` option block, must be formatted as `year YYYY` `month MM day DD at HH:MM`, with `YYYY` the year, `MM` the month, `DD` the day, `HH` the hour and `MM` the minute when the simulation should start.

3.3.2 Reading NetCDF times

When reading a NetCDF input file, Ichthyop will determine the units in which NetCDF time is stored in this file. These units must meet the CF Metadata Conventions and therefore be provided as follows:

```
UNITS since YYYY-MM-DD HH:MM:SS
```

with `UNITS` the units in which the time is stored (usually `seconds`, `days` or `hours`), `YYYY` the year, `MM` the month, `DD` the day, `HH` the hour, `MM` the minutes and `SS` the second of the reference date.

If a NetCDF `time::units` attribute is defined, Ichthyop will try to infer the NetCDF reference date and time units using the CF conventions.

If it fails (i.e. the `time::units` attribute does not follow the CF conventions) or if no `time::units` attribute is provided in the input file, Ichthyop will read the `time_origin` parameter from the `app.time` option block, which must be defined following the CF conventions.

> Caution
>
> When reading two different NetCDF input files (for example one providing the ocean currents and one provinding surface winds), if none meets the CF convention, Ichthyop will apply the units defined in the `time_origin` parameter of Ichthyop configuration file to both, even though they may have different time units. **Therefore, in this case, it is strongly recommended to include CF-like time units attributes manually to each file (cf. below).**

This can be done by using the ncatted command (**Linux users**):

```bash
#!/bin/bash
for file in *nc
do
    ncatted -O -a units,time,o,s,"seconds since 1900-01-01 00:00:00" $file
done
```

This can also be done by using Python as follows:

```python
from glob import glob
import xarray as xr

units = 'seconds since 1900-01-01 00:00:00'
```

```python
time = 'time'

filelist = glob('*nc')

for f in filelist:

    data = xr.open_dataset(f)
    data[time].attrs['units'] = units
    data[time]

    data.to_netcdf(f)
```

> **Note**
>
> The user must replace `time` by the name of the time variable which is used by Ichthyop. Common values are `time`, `scrum_time`, `time_counter`, `ocean_time`.
> The units value must also be chosen consistently with the dataset

4 FAQ

4.1 Java Heap Space

In case of memory issues, run Ichthyop in the Terminal as follows:

```
java -jar -Xms1g -Xmx16g ichthyop_X.Y.Z.jar
```

The first argument provides the initial heap memory while the second argument provides the maximum size of the heap memory. Replace `16g` by any number.

> ❗ Important
>
> You need to make sure that your computer has at least the memory indicated by the `Xmx` parameter

4.2 What to do when Ichthyop does not manage my ocean dataset

Ichthyop developers cannot manage all the different ocean datasets that exist. First, there are too many of them which rely on different assumptions, such as grid layout, vertical coordinates, etc.

Therefore, the Ichthyop developers have decided to first focus on the most used datasets, i.e. NEMO, MARS, ROMS and ocean datatasets that are stored on regular grid.

If your model does not belon to the list, one possibility is to do a bit of pre-processing, in order to convert your input files to a regular, depth-based ocean grid. Different tools can help you with that, such as the XESMF Python package.

4.3 What to do when Ichthyop suddenly fails to launch ?

Situation: without apparent reason, Ichthyop fails to launch, either from GUI or command line. It looks like it starts and crashed instantly.

Solution: delete Ichthyop persistence files (the files that store the information to restart your ichthyop session as it was when you last closed it) In Windows environment such files are gathered in a hidden directory ~/AppData/Local. In this folder delete the ichthyop directory of the previmer/ichthyop directory. In Linux or Mac environement delete the ~/.ichthyop folder. Try to run Ichthyop again.

4.4 How to launch Ichthyop from command line ?

First you have to open a command prompt window on your computer.

For Linux or Mac users open a new Terminal (type Terminal in the Application search bar or the Finder if you are unsure how to open a terminal).

For Windows users, click on the start button > All programs > Accessories > Command Prompt (read more)

From the command prompt windows you need to change the current directory (by default your home directory) to the Ichthyop directory. For instance

```
cd projects/ichthyop/ichthyop-3.2
or
cd Mes\ Documents/Ichthyop/ichthyop-3.2
```

Assuming that in the first case that your ichthyop folder is in directory projects/ichthyop/ichthyop-3.2 and in the second case in directory MesDocuments/Ichthyop/ichthyop-3.2.

List the folder content to check that you are in the right directory : type dir in Widows and ls in Linux or Mac environment.

Launch Ichthyop, with UI, from command line:

```
java -jar ichthyop-3.2.jar
```

Launch Ichthyop, without UI, from command line:

```
java -jar ichthyop-3.2.jar cfg/your_configuration_file.xml
```

With your_configuration_file.xml the name of the XML configuration files that you first created from the UI and that you saved in the cfg/ folder.

4.5 What are the different coastline behaviours in Ichthyop?

First you may want to read {numref}`spatial-int`

Coastline behaviour manages what must be done in the event that the move of a particle takes it inland (which might happen because the simulation time step is not small enough or because there is some additional movement, such as diffusion or swimming, etc.)

Ichthyop offers four different behaviours at coastline:

- NONE: Ichthyop ignores the fact that it is land and just carries on moving the particle around.
- BEACHING: Ichthyop does move the particle inland but "kill" it. From now onward the particle is out of the simulation.
- BOUNCING: the coastline acts as a billard edge and the particle will bounce as a billard ball in the events that the move would take it beyond the coastline. The particle bounces back as much as it would penetrate inland.
- STANDSTILL: the particle gives up on the move that would take it inland and just wait until next time step for trying an other move.

(spatial-int)=

4.6 How does spatial interpolation work in Ichthyop?

A particle in Ichthyop only knows about its environment what is provided by the outputs of the hydrodynamic model. Such information, for instance the zonal and meridional current velocities, are usually provided on an Arakawa C-grid, every U-point and V-point respectively.

Here is the 2D scheme of cells (i, j) bottom left, (i+1, j) bottom right, (i, j+1) top left and (i+1, j+1) top right.

```
    +----V(i, j+1)----+---V(i+1, j+1)---+
    |                 |                 |
    |                 |                 |
U(i-1, j+1)  +      U(i, j+1)    +     U(i+1, j+1)
    |                 |                 |
    |                 |                 |
    +-----V(i, j)-----+----V(i+1, j)----+
    |                 |                 |
    |            *X   |                 |
U(i-1, j)    +      U(i, j)    +     U(i+1, j)
    |                 |                 |
    |                 |                 |
    +----V(i, j-1)----+---V(i+1, j-1)---+
```

Figure 4.1: Particle current location at `X(x, y, z)`

Let's see how the interpolation works for both zonal and meridional velocities. The question we ask is what is the value of U and V at particle location?

We have i=round(x), j=truncate(y) and k=truncate(z), dx=x-i, dy=y-j, dz=z-k

Let's call t, the current time of the simulation, and t0 and t1 the values of the time NetCDF variable bounding t: t0 <= t < t1

We first interpolate the model velocity field at t0:

```
U(t0, x, y, z) = U(t0, i-1, j,   k)   * |(0.5-dx) * (1-dy) * (1-dz)|
               + U(t0, i-1, j,   k+1) * |(0.5-dx) * (1-dy) * dz    |
               + U(t0, i-1, j+1, k)   * |(0.5-dx) * dy     * (1-dz)|
               + U(t0, i-1, j+1, k+1) * |(0.5-dx) * dy     * dz    |
               + U(t0, i,   j,   k)   * |(0.5+dx) * (1-dy) * (1-dz)|
               + U(t0, i,   j,   k+1) * |(0.5+dx) * (1-dy) * dz    |
               + U(t0, i,   j+1, k)   * |(0.5+dx) * dy     * (1-dz)|
               + U(t0, i,   j+1, k+1) * |(0.5+dx) * dy     * dz    |
```

This large expression can be narrowed down to:

```
U(t0, x, y, z) = SUM( U(t0, i+ii-1, j+jj, k+kk) * |(0.5-ii-dx) * (1-jj-dy) * (1-kk-dz)| , ii in [0,1], j
```

with i=round(x), j=truncate(y), k=truncate(z), dx=x-i, dy=y-j, dz=z-k

Similarly, the meridional velocity can be expressed as:

```
V(t0, x, y, z) = SUM( U(t0, i+ii, j+jj-1, k+kk) * |(1-ii-dx) * (0.5-jj-dy) * (1-kk-dz)| , ii in [0:1], j
```

with i=truncate(x), j=round(y), k=truncate(z), dx=x-i, dy=y-j, dz=z-k

It means that the velocity, either zonal or meridional, at particle location is the result of a trilinear interpolation of the height (four above, four below) surrounding velocities in the grid.

Same with U(t1, x, y, z) and V(t1, x, y, z)

Let's take frac = (t - t0) / (t1 - t0). Then we have

```
U(t, x, y, z) = (1 - frac) * U(t0, x, y, z) + frac * U(t1, x, y, z)
```

```
V(t, x, y, z) = (1 - frac) * V(t0, x, y, z) + frac * V(t1, x, y, z)
```

This is the general case when all the surrounding cells are in water. Now what happened if the particle is in a cell adjacent to the coast? Let's say that in our example cell(i+1,j) and cell(i+1, j+1) are land. Basically the interpolation is limited to the four (two above and two below) closest surrounding velocity points:

```
U(t0, x, y, z) = SUM( U(t0, i+ii-1, j, k+kk) * |(0.5-ii-dx) * (1-dy) * (1-kk-dz)| , ii in [0,1], kk in [
```

```
V(t0, x, y, z) = SUM( U(t0, i, j+jj-1, k+kk) * |(1-dx) * (0.5-jj-dy) * (1-kk-dz)| , jj in [0:1], kk in [
```

In order to determine whether a particle is close to the coastline, Ichthyop proceeds in two steps: it first determines in which quater of the cell the grid point is located. Then it checks wether or not the three adjacent cells to this quater are in water.

```
-+------------------+-
 |           *X2    |
 | *X1              |
 |        +(i,j)    |
 |                  |
 |   *X4    *X3     |
-+------------------+-
```

X1 will be considered as “close to coast” if any of cells (i,j+1) (i-1,j) (i-1,j+1) is on land.

X2 will be considered as “close to coast” if any of cells (i,j+1) (i+1,j) (i+1,j+1) is on land.

X3 will be considered as “close to coast” if any of cells (i,j-1) (i+1,j) (i+1,j-1) is on land.

X4 will be considered as “close to coast” if any of cells (i,j-1) (i-1,j) (i-1,j-1) is on land.

4.7 How does Ichthyop manage time ?

Time management is tricky to handle because on the computer side a given time is usually expressed as a number of seconds elapsed since a time origin (e.g. 13629116520 seconds elapsed between 1900/01/01 00:00 and 2014/09/04 09:42), whereas the user expects to read time in a human readable format (e.g. 2014/09/04 09:42).

Ichthyop is no different: the program itself only understands a time as a number of seconds elapsed since a time origin, just like the hydrodynamic datasets ROMS, MARS, NEMO, etc. and time displayed in the console or the GUI uses a human readable format. Since time in the hydrodynamoc dataset is expressed as a number of seconds elapsed since a time origin, Ichthyop must be able to convert a human readable time (for example the time of begining of the simulation) into a number of seconds, so that it can compare this given time value to the time vector of the hydrodynamic dataset and interpolate the velocity fields at the correct time step. The key issue is how to convert a human readable time into a number of seconds elapsed since a time origin ?

In order to do so, we need a calendar that basically details how many days (a day is always considered as a 24h period) are there in each month for each year. Since Ichthyop has to read some variables from the hydrodynamic dataset at a given time, we must make sure that Ichthyop uses the same calendar than the hydrodynamic dataset.

The default calendar used by Ichthyop is the Gregorian calendar (the most widely used civil calendar), the one we use for our daily life. The time origin is set by default at 1900/01/01 00:00. This value can be changed in the configuration file, in the {guilabel}`Time section`: tick the {guilabel}`Show hidden parameters` checkbox, change parameter {guilabel}`Type of calendar` to *Gregorian calendar* and adjust the value of parameter {guilabel}`Origin of time` so that it matches the origin of time set in the hydrodynamic dataset. Such information usually comes as an attribute of the time variable in the NetCDF output files of the hydrodynamic dataset.

Nonetheless some hydrodynamic simulations run with a different calendar than the Gregorian calendar. So far, Ichthyop includes an other calendar that we called the *Climatology calendar*. It is a commonly used calendar for climatological simulations, a year is divided in *12 months of 30 days each*. In order to select this calendar from the editor of configuration, go to the {guilabel}`Time section`, tick the {guilabel}`Show hidden parameters` checkbox and select the *Climatology calendar* for parameter {guilabel}`Type of calendar`. The origin of time for the climatology calendar is set at 01/01/01 00:00 and cannot be changed.

Let's sum up the steps involved in the time management, using the example of the time of beginning of the simulation:

- user provides a time for the beginning of the simulation 2014/09/04 09:42 ;
- user sets up the calendar to be used in Ichthyop, the same one that has been used in the hydrodynamic dataset, e.g. Gregorian calendar with origin of time 1900/01/01 00:00 ;
- Ichthyop converts 2014/09/04 09:42 into a number of seconds using the user-defined calendar, 13629116520 seconds ;
- Ichthyop scans the time variable of the hydrodynamic dataset and identifies that time value 13629116520 falls in between time step 5 and 6 of the hydrodynamic time step (time step 5 and 6 are just an example) ;
- Ichthyop can perfom the time integration of the velocity fields between time steps 5 and 6 of the hydrodynamic dataset and starts advecting the particles.

We provide a very simple utiliy programs in the Time converter repository, that illustrates how Ichthyop performs the time conversion, given a calendar and a time of origin.

Last bu not least: what if your hydrodynamic dataset uses an other calendar than Gregorian or Climatology calendars? Two options:

1. Contact the developpers and ask how much work would it be to include your calendar in Ichthyop ?
2. Select an existing calendar that is the most similar to yours and trick Ichthyop by providing a human readable time that you know it will be converted in the correct time value for the hydrodynamic dataset (thanks to the **Time converter** utility program).

4.8 How does Ichthyop interpolate the hydrodynamic dataset in time ?

Let's say that the hydrodynamics output dataset is archived with a 5 days time step and Ichthyop runs with a 1 hour time step.

Let's call tn a given time index in the hydrodynamic dataset and tnp1 the following one. And let's call Tr a variable from the dataset (either current velocity, temperature, free surface elevation, etc.). Let's call time the time variable of the hydrodynamic dataset, always expressed in seconds elapsed from a given origin.

Ichthyop does a linear interpolation to estimate the value of Tr at any given time between time(tn) and time(tnp1). For t, a given time index such as time(t) >= time(tn) and time(t) < time(tnp1), we have :

```
Tr(t) = (1 - x) * Tr(tn) + x * Tr(tnp1)
```

with x `= ( time(t) - time(tn) ) / ( time(tnp1) - time(tn) )`

4.9 Why do I get warning "CFL broken for W 1.208" ?

CFL stands for Courant–Friedrichs–Lewy condition. It is a necessary condition for stability while solving the equation of movement in Ichthyop.

We want at all time `(U * dt) / dX` stricly inferior to 1 with U the velocity (vertical in this specific case) and dX the move. **The warning informs you that the CFL condition has been broken for the vertical velocity and that it might jeopardize the numerical stability of the model.**

Try to decrease the time step (dt) in your configuration file (Time > Time step), let's say divide it by two. As explained in the configuration editor, an acceptable estimation for dt could be dt = 0.7 * dGrid / Umax with dGrid the average length of the grid cells and Umax the order of magnitude of the fastest current velocities in the hydrodynamic model (locally and punctually the vertical velocities can be intense). Nonetheless the smaller the better and as long as decreasing the time step does not slow down too much your simulations, you should always go for a smaller value that the previous estimation.

Part II

Documentation

5 Particle release

In the present section, the different release processes that are implemented within Ichthyop are described. The parameters that are associated with the release processes must be included within `release` blocks (cf. Section 3.1).

5.1 Stain release

`Main - Release Stain`

The release stain consists in releasing particles within a given disk. The user provides the longitude and latitude of this disk center and its radius (in *m*), and the number of particles to release.

For a 3D simulation, the release stain becomes a cylinder and the user also has to provide the depth of the center of the cylinder and its height (both in *m*).

Particles are randomly distributed within a release stain (Figure 5.1)

Figure 5.1: Example of a stain release. Note that here the disk is truncated due to the coast.

5.2 Zone release

`Advanced - Release - From zones`

The zone release method allows to release particles in user-defined areas (see Section 2.1.3 and Section 3.2)

By default, the number of particles released in each zone (N_k) is equal to

$$N_k = N_{tot} \times \frac{S_k}{\sum_{i=1}^{N} S_i}$$

with N_{tot} the total number of released particles, S_k the surface of the k^{th} release area and N the number of release areas.

> ⚠ Warning
>
> The thickness of the area is not taken into account when computing the proportion of particles that will be released, only the horizontal surface

However, the user has the possibility to control the proportion of particles that is released in each zone.

For details on how to define release zones, see Section 2.1.3 and Section 3.2

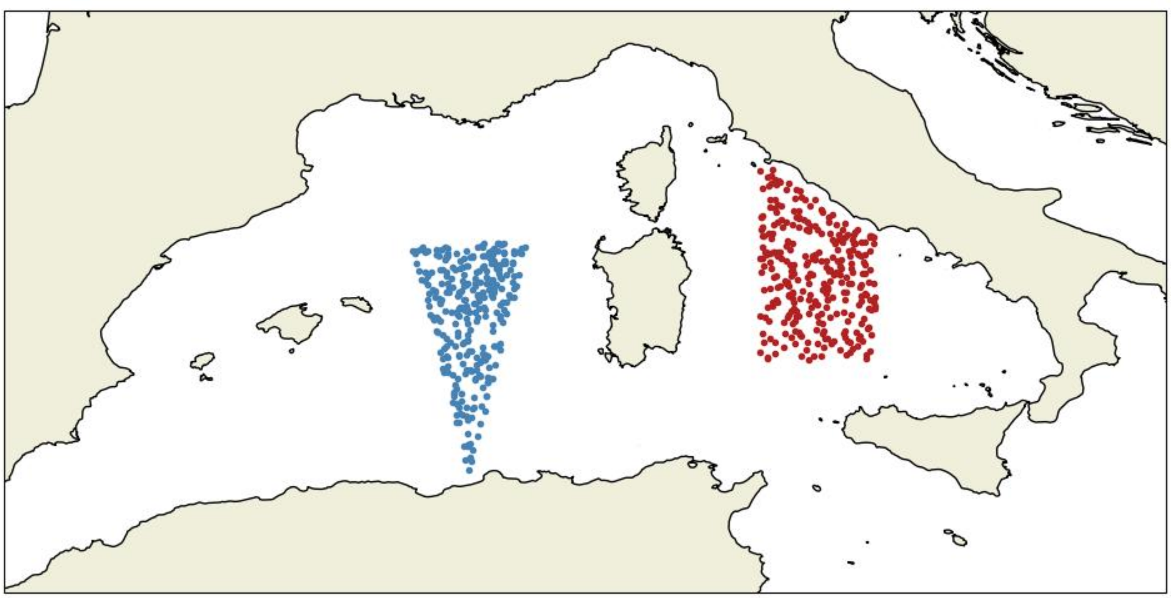


Figure 5.2: Example of a zone release.

5.3 Text release

`Advanced - Release - Drifters from text file`

Particles can also be released by providing a text file containing their release coordinates.

The file must be formatted as follows:

```
# 3D simulations
# longitude latitude depth
-5.45 48.30 -5
-5.45 48.30 -10
-5.45 48.30 -15
-5.45 48.30 -20
```

Each line is a particle. Each character starting with `#` is considered as a comment. If the depth column is not provided, depth will be set to `0`.

For 2D simulations, the depth column will be ignored.

> **Caution**
>
> Note that the columns must be separated by spaces.

5.4 Patchy release

`Advanced - Release - Patches in zones`

It is a combination of the stain release and zone release methods.

The patchy release method (`PatchyRelease.java`) can be viewed as an extension of the release stain method, where several stains are randomly created.

The number of stains and of particles per stain are provided. The radius and thickness (if 3D simulation) of each patch is also provided

Additionally, there is the possibility to release patches in each release zone. This is achieved by setting the `per_zone` parameter to true. An example is provided in Figure 5.3.

If this parameter is set to false, the bounding box around the defined release zones (if any) or the full domain is used to release the patches, as shown in Figure 5.4.

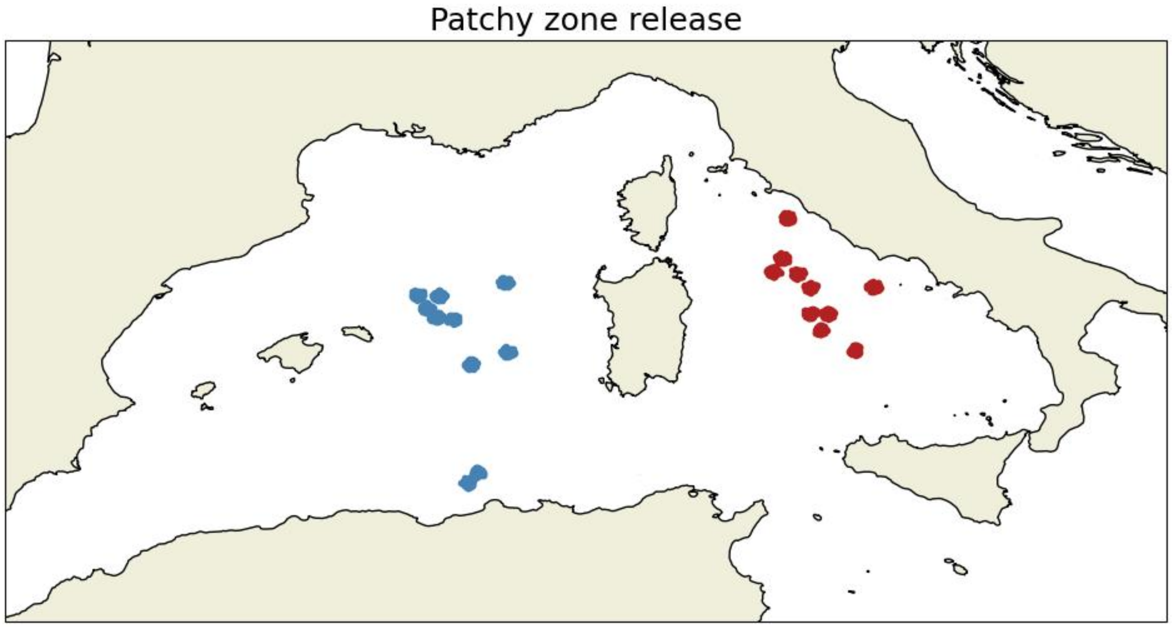


Figure 5.3: Example of a patchy zone release.

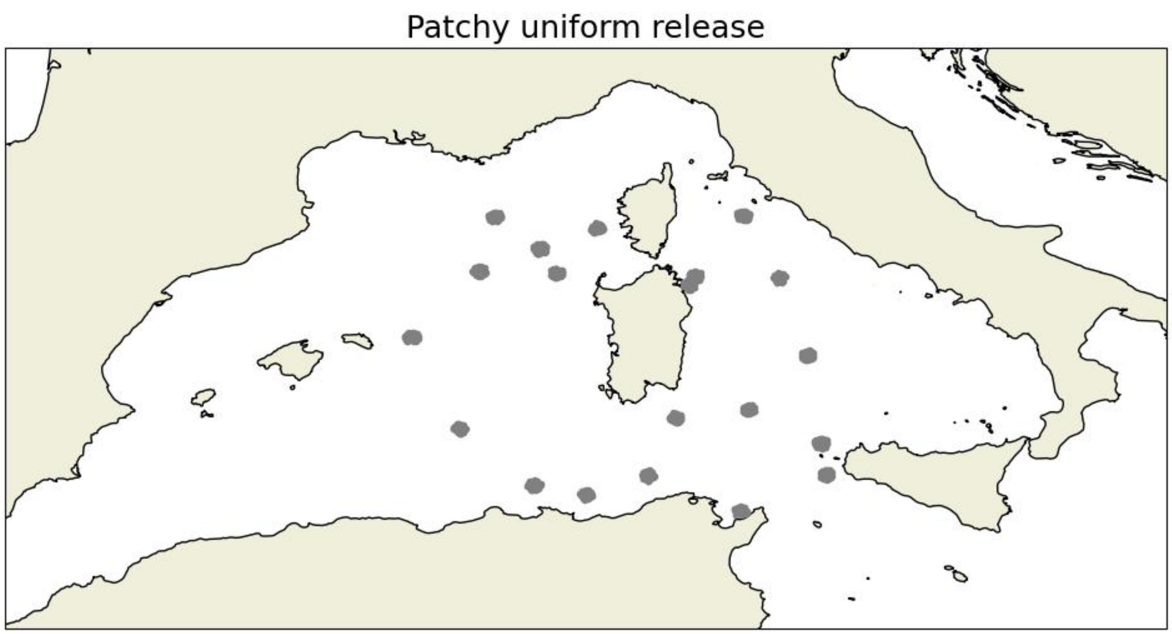


Figure 5.4: Example of a patchy uniform release.

5.5 Surface and bottom releases

`Advanced - Release - At bottom`

The surface and bottom release methods allow to randomly release particles over the entire domain, either at the surface or at the bottom (for 3D simulations only).

The only parameter required by these methods is the number of particles. There is no control over the area where the particles are released.

An example of a surface release is shown in Figure 5.5.

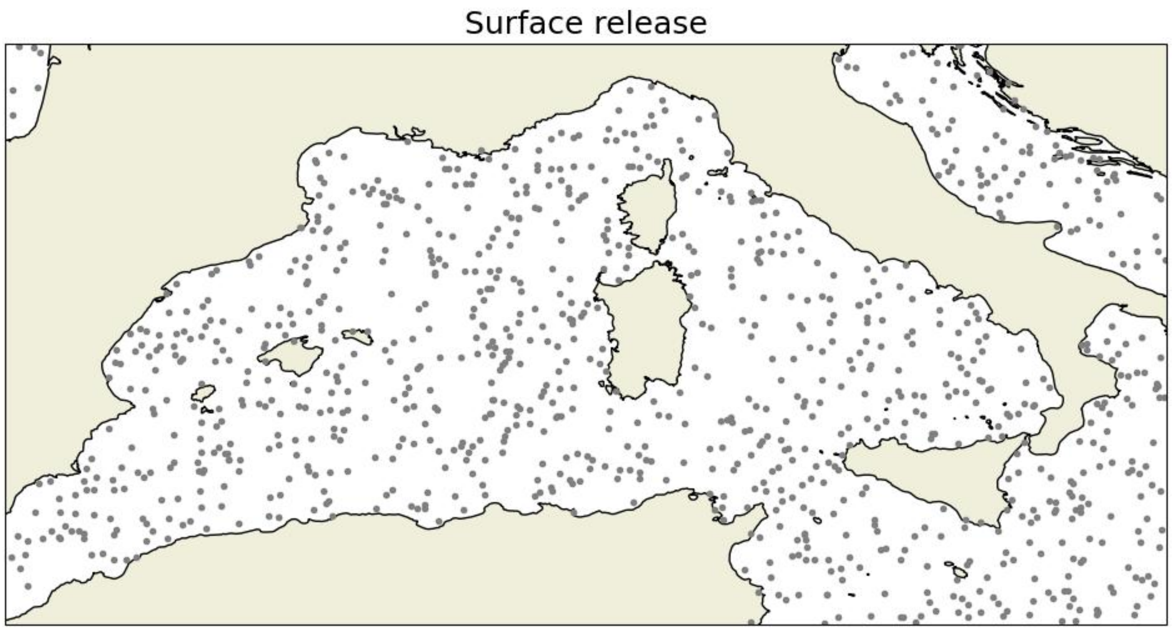


Figure 5.5: Example of a surface release.

5.6 Netcdf release

`Advanced - Release - Restart model`

Ichthyop also offers the possibility to use an Ichthyop output file as a simulation starting point. The location of the NetCDF file to use as a restart must be provided.

6 Release schedule

`Advanced - Release - Schedule`

There is also the possibility to schedule several release events as a list of release date strings, formatted as `year YYYY month MM day DD at YY:MM`.

The resulting output file will contain the same number of time-steps as in the case of a simple release, but the number of particles will be multiplied by the number of release events.

> **Tip**
>
> It is recommended to avoid using the release schedule feature and instead to change the value of the `initial_time` parameter (Section 3.3.1)

> **Warning**
>
> If the simulation duration to 30 days `initial_time` is set to `year 2000 month 01 day 01 at 00:00` and your first release event is set to `year 2001 month 01 day 01 at 00:00`, Ichthyop will run without without any particles from 2000-01-01 to 2001-01-01

> **Warning**
>
> If the simulation duration to 30 days `initial_time` is set to `year 2000 month 01 day 01 at 00:00` and your first release event is set to `year 2001 month 01 day 01 at 00:00`, the final time-step will occurs in 2001-01-30.

7 Ichthyop processes

In this section, the different Ichthyop processes are described. These processes are implemented in the classes within the `action` folder.

Parameters associated with these processes must be included in `action` blocks. They can be activated or deactivated by setting the `enabled` tag (cf. Section 3.1).

7.1 Activation/Deactivation of a process

Each process can be enabled or disabled by ticking on the `enable` tick box.

Furthermore, it is also possible to activate/deactivate some processes depending on the particle age (in days) or size (in *cm*).

This is done by providing, for each of these processes, the following parameters:

- `activation_variable`, which indicates whether age (in days) or length (in cm) must be used
- `activation_minimum_class_value`, which indicates the age (in days) or lengh (in cm) when the process becomes active. **Default is 0**
- `activation_maximum_class_value`, which indicates the age (in days) or lengh (in cm) when the process becomes inactive. **Default is ∞**

> Caution
>
> The length can be used only if one of the different growth process (see Section 7.3.8) is enabled

7.2 Transport processes

7.2.1 Advection

The primary process in Ichthyop is the advection by the ocean currents.

7.2.2 Horizontal advection

The particle grid coordinates will change according as follows:

$$X(t + \Delta t) = X(t) + \frac{U \times \Delta t}{\Delta X}$$

$$Y(t + \Delta t) = Y(t) + \frac{U \times \Delta t}{\Delta Y}$$

with X, Y the particle grid position, t the time, Δt the time step (in seconds), U and V the zonal and meridional velocities (in $m.s^{-1}$) and ΔX and ΔY the zonal and meridional extent of the grid cell.

7.2.3 Vertical advection

For 3D simulations, vertical advection can also be activated. Particle depth, in grid coordinates, is given by:

$$Z(t + \Delta t) = Z(t) + \frac{W \times \Delta t}{\Delta Z}$$

with Z the particle vertical grid position, t the time, Δt the time step (in seconds), W the vertical velocity (in $m.s^{-1}$) and ΔZ the height of the grid cell.

7.2.4 Buoyancy

A buoyancy module allows to displace a particle vertically, depending on it's density and the sea water density, following Parada et al. (2003).

The buoyancy-induced vertical velocity of the particle is given by:

$$W_{buoy} = \frac{1}{24} \times g \times a \times b \times \frac{\rho_{water} - \rho_{part}}{\rho_{water}} \mu^{-1} \log\left(2\frac{a}{b} + 0.5\right);$$

with a and b the semi-major axis and semi-minor axis of an ellipse (`mean_major_axis` and `mean_minor_axis` parameters), μ the molecular viscosity (`molecular_viscosity` parameter), ρ_{water} the water density, ρ_{part} the particle density and g the gravitational acceleration ($cm.s^{-2}$).

There are two possibilities to define the particle buoyancy, which are provided by the `density.method` parameter:

- If `constant`, then a constant density (in $g.cm^{-3}$) is used, which is provided by the `particle_density` parameter.

- If `file`, then the buoyancy is read from a file and can vary with either age or length. The class that is used is provided by the `density.class` parameter.

The density CSV file must be formatted in the following way:

```
Age(Days);Density (g/cm3)
0;1.0235625
1;1.02374
2;1.023335
2.5;1.0239
2.75;1.025672
```

7.2.5 Daily vertical migration

Daily migration of the particles can be activated by providing daytime and nighttime depths, and the timing of the sunset and sunrise.

If the `daytime_depth_file` parameter is defined, it provides the depth of the particle during daytime. It is formatted as follows:

```
Age (day);Depth (m)
0.0;-20
3.0;-25
5.0;-30
8.0;-35
```

If this parameter is not found, a constant daytime depth, provided by the `daytime_depth` parameter, is assumed.

Same thing for the nighttime depths, which can be set by either the `nighttime_depth_file` or the `nighttime_depth` parameters.

The sunset and sunrise hours are set by the `sunset` and `sunrise` parameters, which must have a `HH::mm` format.

> Warning
>
> When the target depth is greater than the total depth, the particle does not move.

7.2.6 Horizontal dispersion

Horizontal dispersion is implemented following Peliz et al. (2007). The horizontal velocity component due to diffusion is:

$$u_T = \delta\sqrt{\frac{2K_h}{\Delta t}}$$

with Δt the time step, δ a random number between -1 and 1 and K_h the imposed explicit Lagrangian horizontal diffusion.

The value of K_h depends on the mean mesh size where the particle is located:

$$K_h = \epsilon^{1/3} l^{4/3}$$

with l is the mean mesh size and ϵ is the turbulent dissipation rate (in $m^2.s^{-3}$).

In the Ichythop implementation (`HDispAction.java`), the displacement is computed:

$$\Delta X = u_T \times \Delta t$$

Combining all together, we obtain the following equation:

$$\Delta X = \delta \times \sqrt{\frac{2\epsilon^{1/3} l^{4/3}}{\Delta t}} \times \Delta t$$

By putting the rightmost Δt within the square root, and by putting the l and ϵ terms outside, this expression can be simplified as follows:

$$\Delta X = \delta \times \sqrt{2\Delta t} \times \epsilon^{1/6} \times l^{2/3}$$

The value of l is computing by averaging the mesh size along the X and Y dimensions.

7.2.7 Ontogenetic vertical migration

The ontogenetic migration module controls the migration of particles at different depths as a function of age. Its implementation in Ichthyop follows the CMS one. It uses a CMS configuration file, which is formatted as follows:

```
nTime
nDepth
z1 z2 z3 z4 z5 z6 ... znDepth
t1 t2 t3 t4 t5 t6 ... tnTime

P1,1 P1,2 ... P1,nTime
P2,1 P2,2 ... P2,nTime
P3,1 P3,2 ... P3,nTime
          ...
PnDepth,1 PnDepth,2 ... PnDepth,nTime
```

The first line provides the number of time steps in the CMS file, the second line provides the number of vertical levels in the CMS file. The third line provides the depth values and the fourth lines provides the time step values.

The remaining lines provide the probability matrix $P_{z,t}$.

> **Note**
>
> The sum of the probability matrix along the depth dimension should equal 100.

At each time step, the index of the CMS time step, k, is determined by comparing the simulation and CMS times as follows:

$$t_{CMS}(k) < t \leq t_{CMS}(k+1)$$

When the CMS time index changes, all the particles are randomly distributed on the vertical, following the probability distribution of the given CMS time. If the CMS time index remains unchanged, nothing is done.

7.2.8 Wind drift

The wind-drift is implemented in the `WindDriftFileAction.java` class. This class requires that 2D (time, latitude, longitude) wind fields are provided. This is done by providing a `input_path` and a `file_filter` parameters, which specify the location and names of the wind files.

The user also provides `field_time`, `wind_u`, `wind_v`, `longitude`, `latitude` parameters, which specify the names of the time, zonal wind, meridional wind, longitude and latitude variables in the NetCDF wind files.

The user also provides a `depth_application` parameter, which specifies the depth at which the wind will impact the trajectories (only valid for 3D simulations), a `wind_factor` (F, multiplication factor), an `angle` (θ) and a `wind_convention` parameter (ε, +1 if convention is ocean-based, i.e. wind-to, -1 if convention is atmospheric based, i.e. wind-from).

The changes in particle longitude (λ) and latitude (ϕ) is computed as follows:

$$\Delta\lambda_0 = \frac{\Delta t \times U_W}{\frac{R\pi}{180} \times \cos(\frac{\pi\phi}{180})}$$

$$\Delta\phi_0 = \frac{\Delta t \times V_W}{\frac{R\pi}{180}}$$

$$\Delta\lambda = \varepsilon \times F \times \left(\Delta\lambda_0 \times \cos\left(\frac{\theta\pi}{180}\right) - \Delta\phi_0 \times \sin\left(\frac{\theta\pi}{180}\right)\right)$$

$$\Delta\phi = \varepsilon \times F \times \left(\Delta\lambda_0 \times \sin\left(\frac{\theta\pi}{180}\right) + \Delta\phi_0 \times \sin\left(\frac{\theta\pi}{180}\right)\right)$$

where $\Delta\lambda_0$ and $⊠⊠_{0} $ are the changes in longitudes and latitudes due to wind, $\Delta\lambda$ and $⊠⊠$ are the effective changes in longitudes and latitudes of the particle when taking into account the angle θ, Δt is Ichthyop time-step (in days) and U_w and V_w are the zonal and meridional wind components (in $m.s^{-1}$).

> **Note**
>
> R is the Earth Radius and equals 6367.74 km. In the code, $\frac{R\pi}{180}$, which is the distance in m of a 1 degree cell, is approximated to 111138 m

7.2.9 Random swimming

Ichthyop allows particles to randomly swim. This is achieved by using the `SwimmingAction.java`. The velocity may vary with the age of the particle.

The user provides a CSV absolute velocity file (`velocity_file` parameter), which must me formatted as follows:

```
Age (days);Speed (m/s)
0;0.1
5;0.2
15;0.3
```

The user also provides a boolean parameter (`constant_velocity`) that specifies whether the velocity should remain as defined in the CSV file, or if the absolute velocity should be randomly selected as follows:

$$\|U\| = \|U\|_{file} \times \kappa$$

with κ a random value in the $[0, 2]$ interval and $\|U\|_{file}$ the velocity defined in the CSV file.

At each time-step, the zonal and meridonal velocities are defined as follows:

$$U = \kappa' \varepsilon \|U\|$$

$$V = \varepsilon' \sqrt{\|U\|^2 - U^2}$$

with κ' a random value between 0 and 1, and ε and ε' random values either equal to 1 or -1.

7.2.10 Wave drift

Ichthyop can take into account the effects of waves on the particles trajectories, following the Stokes drift equations (Stokes (2009)):

$$U_S = w \times k_{wave} \times a^2 \times \exp\left(2 \times k_{wave} \times z\right)$$

where $w \times k \times a^2$ is the Stokes drift velocity, k_{wave} is the wave number and z is the depth.

In Ichthyop, the user provides the zonal and meridional components of the Stokes drift (U_{stokes} and V_{stokes}), the zonal and meridional wave velocities (U_{wave} and V_{wave}) and the wave period (T_{wave}).

> **!** Important
>
> Note that all the fields provided by the user are 2D!

First, the value of k_{wave} is computed from the wave velocity and the wave period:

$$\|U_{wave}\| = U_{wave}^2 + V_{wave}^2$$

$$\lambda_{wave} = \|U_{wave}\| \times T_{wave}$$

$$k_{wave} = \frac{2\pi}{\lambda_{wave}}$$

Then, the horizontal displacement due to the Stokes drift is computed as follows:

$$\Delta X = F \times U_{stokes} \times \Delta t \times \exp\left(2 \times k_{wave} \times z\right)$$

$$\Delta Y = F \times V_{stokes} \times \Delta t \times \exp\left(2 \times k_{wave} \times z\right)$$

with k_{wave} the wave number as computed above, U_{stokes} and V_{stokes} the zonal and meridional components of the Stokes drift, z the depth and F a multiplication factor provided by the user.

7.2.11 Coastal behaviour

Coastal behaviour defines what is done before aplying the change in location to an Ichthyop particle

7.2.11.1 Default

The default behavior is to do nothing prior the displacement of a particle. The change in coordinate of the particle is applied whatsoever, and the particle may be inland. It remains alive but then its behavior is uncertain.

> ⚠ Warning
>
> It is strongly advised not to use this default mode

7.2.11.2 Beaching

Before the particle is moved, Ichthyop checks whether the coming displacement will bring the particle inland. If so, the particle is moved and then is killed, with the BEACHED mortality status.

7.2.11.3 Standstill

Before the particle is moved, Ichthyop checks whether the coming displacement will bring the particle inland. If so, its deplacement is cancelled and the particle remains at the same position until a new displacement brings it in a water cell.

7.2.11.4 Bouncing

In the bouncing mode, Ichthyop will check whether the displacement will bring the particle inland. If so, the particle will bounce on the coast. First, whether the bouncing occurs on a meridional or a zonal coastline is determined.

In case of a meridional coastline, as shown in Figure 7.1, the calculation of the new position is performed as follows.

Let's assume that the particle is at the position (x, y) and would normally move at the postion $(x + \Delta x, y + \Delta y)$. We suppose that

$$\Delta y = \Delta y_1 + \Delta y_2, \tag{7.1}$$

where Δy_1 is the meridional distance between the particle and the coastline, and Δy_2 is the distance that the particle would travel on land.

In the bouncing mode, the position increment can be written as

$$\Delta y_{cor} = \Delta y_1 - \Delta y_2 \tag{7.2}$$

By replacing Δy_2 using Equation 7.1, we can write:

$$\Delta_{cor} y = \Delta y_1 - (\Delta y - \Delta y_1)$$

$$\Delta_{cor} y = 2\Delta y_1 - \Delta y$$

7.3 Biological processes

7.3.1 Orientation

Active swimming has been implemented in Ichthyop following the work of Romain Chaput. Three active swimming behaviours have been implemented: the rheotaxis orientation (i.e. against the current), the cardinal orientation (i.e toward a given direction) and the reef orientation, i.e. orientation toward points of interests.

These three implementations involve the computation of a swimming velocity and direction. The former is common to all three methods, but the directions depend on the method considered.

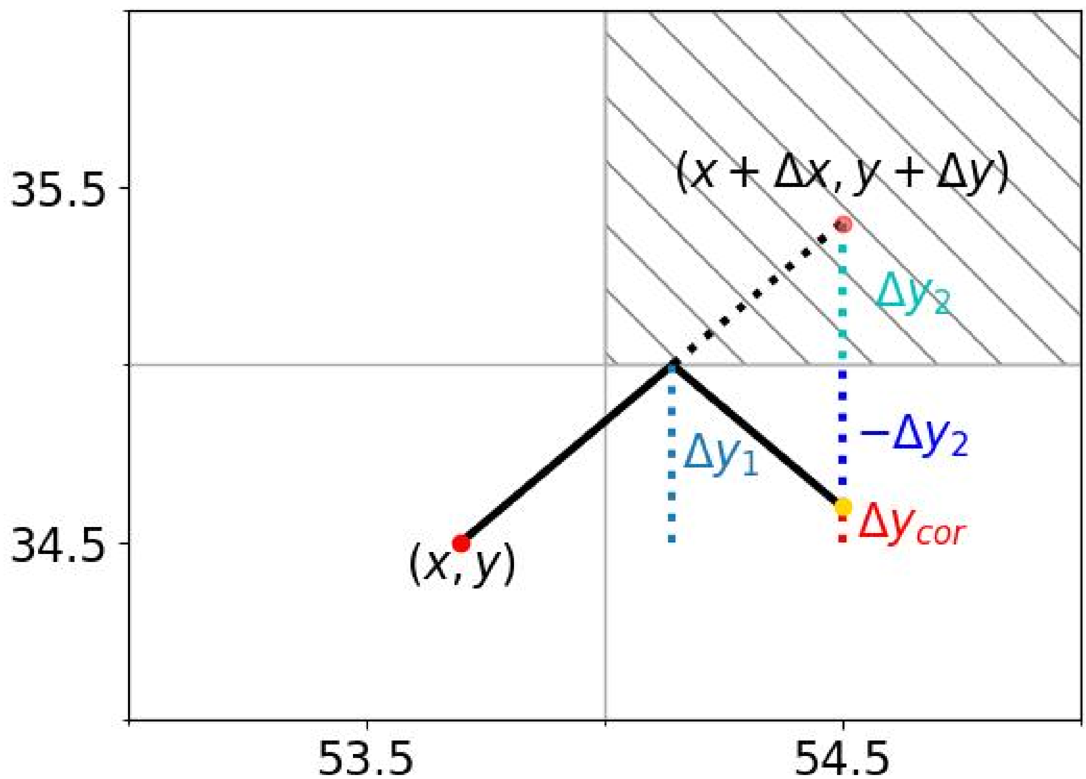


Figure 7.1: Coastal behaviour in bouncing mode.

```python
import os
from glob import glob
import subprocess

pom_file = os.path.join('..', 'pom.xml')

command = ['mvn', 'package', '-f', pom_file, '-DskipTests=true']
subprocess.run(command, capture_output=False, text=False, stdout=subprocess.DEVNULL)

conflist = glob(os.path.join('ichthyop_configs', '*'))
conflist.sort()

confnames = [os.path.basename(f) for f in conflist]
for conf in confnames:
    if os.path.isdir(os.path.join('ichthyop_output', conf)):
        print(f"Configuration {conf} already exists in output directory")
        continue
    else:
        jarname = glob(os.path.join('..', 'target', 'ichthyop*[0-9].jar'))
        print(jarname)
        jarname.sort()
        jarname = jarname[-1]
        print(jarname)
        command = ['java', '-jar', jarname, os.path.join('ichthyop_configs', conf, f'{conf}.xml'
        print(command)
        result = subprocess.run(command, capture_output=False, text=False, stdout=subprocess.DEVN
```

```
Configuration card already exists in output directory
Configuration combined already exists in output directory
Configuration control already exists in output directory
Configuration reef already exists in output directory
Configuration rheo already exists in output directory
```

7.3.2 Swimming velocity

The orientation processes all share common features. They both depend on a swimming velocity and random directions. The methods described below differ on the way the random directions are drafted.

Swimming velocity is computed following Staaterman, Paris, and Helgers (2012):

#|echo: false

$$V = V_{hatch} + (V_{settle} - V_{hatch})^{\log(age)/\log(PLD)}$$

with V_{hatch} and V_{settle} the larval velocity at hatching and settle, A the age of the larva and PLD the transport duraction.

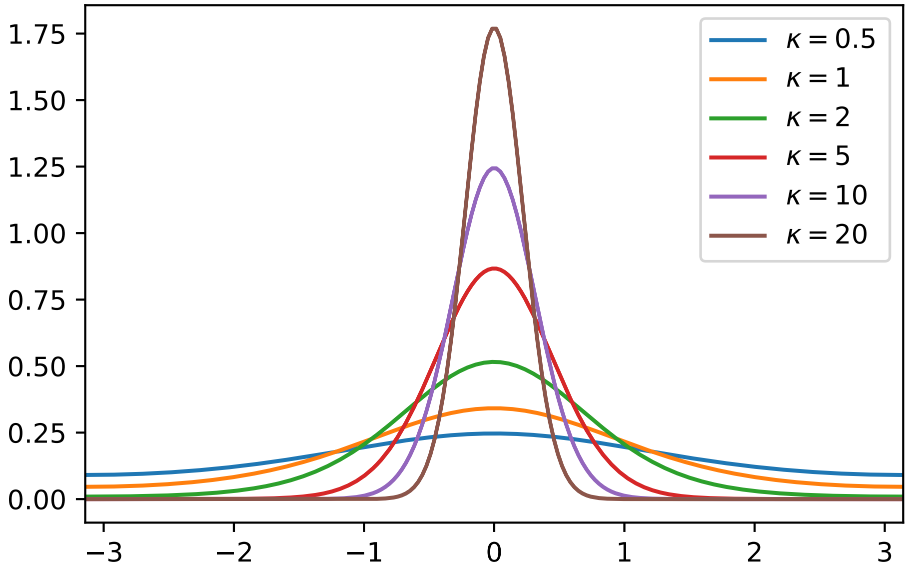
7.3.3 Von Mises distributions

Von Mises distribution is used in all three methods.The Von Mises distribution is given by:

$$f(\theta, \mu, \kappa) = \frac{\exp(\kappa \cos(\theta - \mu))}{2\pi I_0(\kappa)}$$

where $I_0(\kappa)$ is the modified Bessel function of the first kind of order 0, μ is angle where the distribution is centerred and κ is the concentration parameter. The distribution is as follows:

```python
import matplotlib.pyplot as plt
import numpy as np
from scipy.special import i0

x = np.linspace(-np.pi, np.pi, 200)

def von_misses(kappa):
    mu = 0
    y = np.exp(kappa*np.cos(x-mu))/(2*np.pi*i0(kappa))
    return y

plt.figure()

for i in [0.5, 1, 2, 5, 10, 20]:
    plt.plot(x, von_misses(i), label=f'$\kappa = {i}$')
plt.legend()
plt.xlim(x.min(), x.max())
plt.show()
```



For computation purposes, all the Von Mises drafts performed in Ichthyop are done by using $mu = 0$. The angles are thus centerred around 0. Then, the *mu* value is added.

> **Note**
>
> $\theta = 0$ is eastward, $\theta = \frac{\pi}{2}$ is northward, etc.

7.3.4 Computation of displacement

Given a swimming velocity V and a direction θ, the larva displacement (in m) is computed as follows:

$$\Delta X = V \times \cos(\theta) \times \Delta t$$

$$\Delta Y = V \times \sin(\theta) \times \Delta t$$

with Δt the time step. Next, the corresponding change in longitude (λ) and latitude (φ) is computed as follows:

$$\Delta \lambda = \frac{\Delta X}{111138 \times \cos \varphi}$$

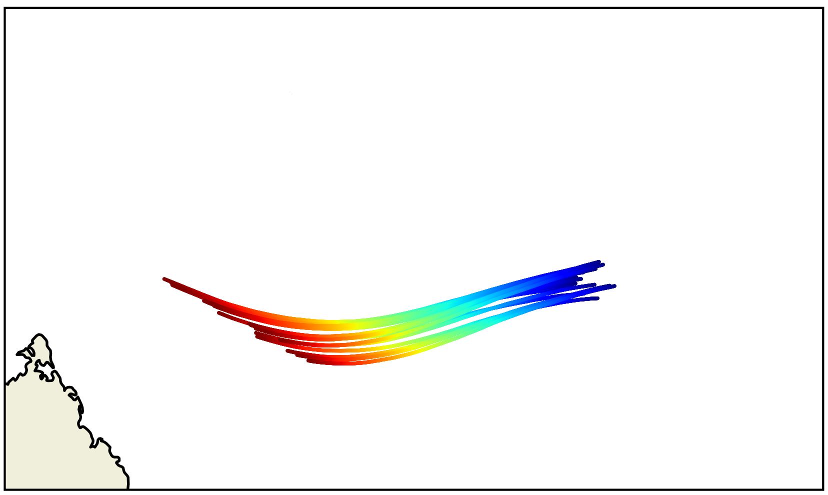
$$\Delta \varphi = \frac{\Delta Y}{111138}$$

7.3.5 Cardinal orientation

In cardinal orientation, the user provides a fixed heading θ_{card} and a fixed κ parameter. Then, at each time step, a new angle is randomly drafted following a Von Misses distribution $f(\theta, \theta_{card}, \kappa)$.

```python
import xarray as xr
import cartopy.crs as ccrs
import cartopy.feature as cfeature
import os
from glob import glob
import matplotlib.pyplot as plt
import numpy as np
import matplotlib as mpl

config = 'card'

if os.path.isdir('ichthyop_output'):
    pattern = os.path.join('ichthyop_output', config, '*nc')
else :
    pattern = os.path.join('..', '..', 'ichthyop_output', config, '*nc')
pattern

filelist = glob(pattern)
filelist

data = xr.open_mfdataset(filelist, decode_times=False)
data

# +
lon = data['lon'].values
lat = data['lat'].values
mort = data['mortality'].values

lon = np.ma.masked_where(mort > 0, lon)
lat = np.ma.masked_where(mort > 0, lat)
ntime, ndrifters = lon.shape

# +
time = np.arange(ntime)
drifters = np.arange(ndrifters)

d2d, t2d = np.meshgrid(drifters, time)
```

```
# +
plt.figure()

projin = ccrs.PlateCarree()
projout = ccrs.PlateCarree()

ax = plt.axes(projection = projout)
ax.scatter(lon[:, :], lat[:, :], c=t2d, marker='.', transform=projin, s=0.5, cmap=mpl.colormaps['
feat = ax.add_feature(cfeature.LAND)
feat = ax.add_feature(cfeature.COASTLINE)
ax.set_extent([49, 55.25, -13.13, -9.45], crs=projin)
```



7.3.6 Rheotaxis orientation

In the rheotaxis orientation method, the particles swim against the current. The user only provides a kappa parameter.

First, the angle of the current is computed as follows:

$$\theta_{current} = \arctan 2(V_{current}, U_{current})$$

Then, the angle that the particle must follow is given by adding π:

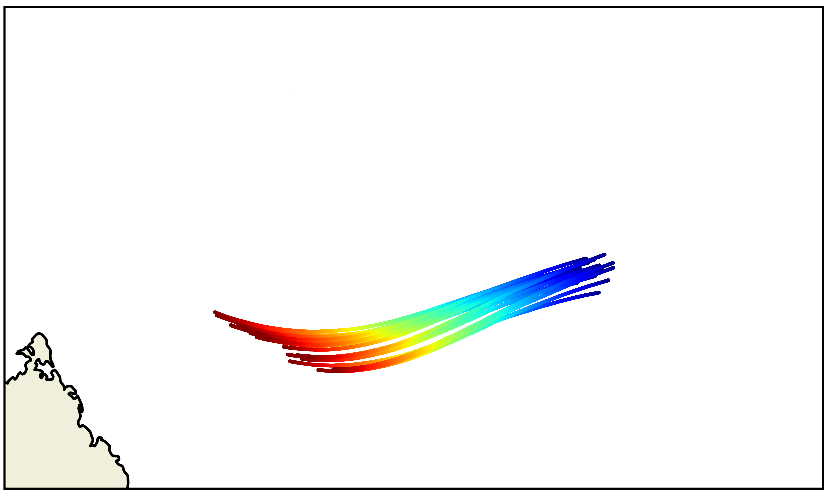
$$\theta_{direction} = \theta_{current} + \pi$$

Finally, a random angle is picked up following a Von Mises distribution $f(\theta, \theta_{direction}, \kappa_{reef})$

(ref-rheo)=

```
import xarray as xr
import cartopy.crs as ccrs
import cartopy.feature as cfeature
import os
from glob import glob
import matplotlib.pyplot as plt
import numpy as np
import matplotlib as mpl

config = 'rheo'

if os.path.isdir('ichthyop_output'):
    pattern = os.path.join('ichthyop_output', config, '*nc')
else :
    pattern = os.path.join('..', '..', 'ichthyop_output', config, '*nc')
pattern

filelist = glob(pattern)
filelist

data = xr.open_mfdataset(filelist, decode_times=False)
data

# +
lon = data['lon'].values
lat = data['lat'].values
mort = data['mortality'].values

lon = np.ma.masked_where(mort > 0, lon)
lat = np.ma.masked_where(mort > 0, lat)
ntime, ndrifters = lon.shape

# +
time = np.arange(ntime)
drifters = np.arange(ndrifters)

d2d, t2d = np.meshgrid(drifters, time)

# +
```

```
plt.figure()

projin = ccrs.PlateCarree()
projout = ccrs.PlateCarree()

ax = plt.axes(projection = projout)
ax.scatter(lon[:, :], lat[:, :], c=t2d, marker='.', transform=projin, s=0.5, cmap=mpl.colormaps['
feat = ax.add_feature(cfeature.LAND)
feat = ax.add_feature(cfeature.COASTLINE)
ax.set_extent([49, 55.25, -13.13, -9.45], crs=projin)
```



7.3.7 Reef orientation

In the reef orientation method, the larva will target the closest target area (for instance coral reef). These areas are defined in an XML zone file by a polygon and a zone-specific κ parameter. The user also provides the sensory detection threshold of the larva (maximum detection distance β).

If the distance between the particle and the barycenter of the closest reef (D) is below the detection thereshold β, the larva will swim in the direction of the reef.

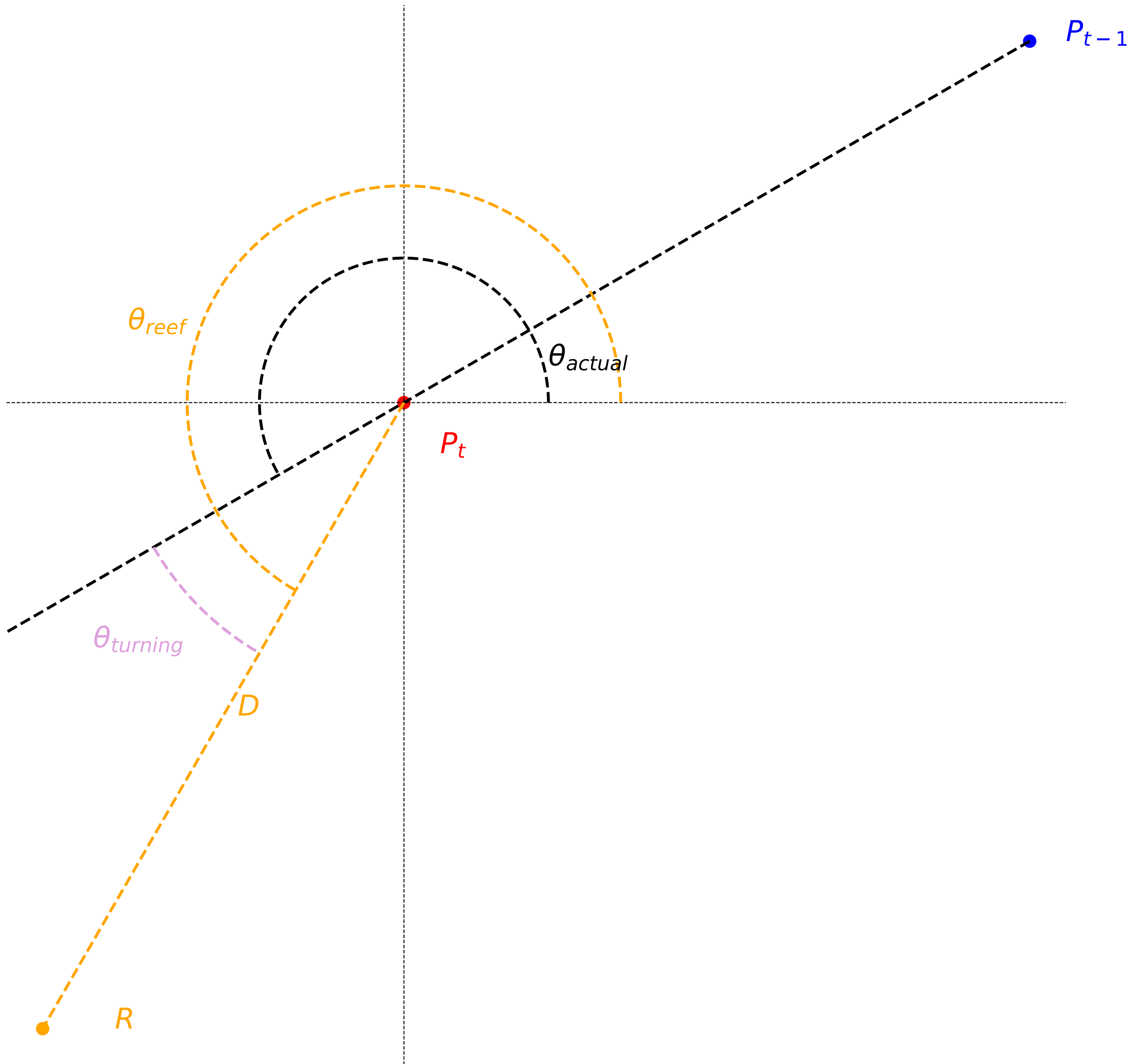
(ref-orientation)=

```
import numpy as np
import matplotlib.pyplot as plt

plt.rcParams['font.size'] = 15
```

```
plt.figure(figsize = (10, 10))

xnew = 0
ynew = 0

#draw point at orgin
plt.plot(xnew, ynew, color = 'red', marker = 'o')
plt.gca().annotate('$P_{t}$', xy=(0 + 0.05, 0 - 0.07), xycoords='data', color='red')

#draw circle
r = 1.5
angles = np.linspace(0 * np.pi, 2 * np.pi, 100 )
xs = r * np.cos(angles)
ys = r * np.sin(angles)
plt.plot(xs, ys, color = 'k', ls='--', lw=0.5)

angle_old = np.pi / 6.
rold = 1.
xold = rold * np.cos(angle_old)
yold = rold * np.sin(angle_old)

plt.plot(xold, yold, marker='o', color='blue')
plt.gca().annotate('$P_{t - 1}$', xy=(xold + 0.05, yold), xycoords='data', color='blue')

angle_reef = np.pi + np.pi / 3.
print(np.rad2deg(angle_reef))
rreef = 1
xreef = rreef * np.cos(angle_reef)
yreef = rreef * np.sin(angle_reef)
print(yreef)

plt.plot(xreef, yreef, marker='o', color='orange')
plt.gca().annotate('$R$', xy=(xreef + 0.1, yreef), xycoords='data', color='orange')

plt.gca().annotate(r'$D$', xy=(0.5*(xreef) + 0.02, 0.5*yreef), xycoords='data', color='orange', h

plt.axvline(xnew, color='k', ls='--', lw=0.5)
plt.axhline(ynew, color='k', ls='--', lw=0.5)

p = np.polyfit([xold, xnew], [yold, ynew], deg=1)
xtemp = np.linspace(xold, xold -2, 100)
plt.plot(xtemp, np.polyval(p, xtemp), color='black', ls='--')
```

```
angle_current = angle_old + np.pi
tmp_angle = np.linspace(0, angle_current, 100)
rtmp = 0.2
plt.plot(0 + rtmp * np.cos(tmp_angle), 0 + rtmp * np.sin(tmp_angle), color='k', ls='--')
plt.gca().annotate(r'$\theta_{actual}$', xy=(xnew + 0.2, ynew+0.05), xycoords='data', color='k')


plt.plot([xnew, xreef], [ynew, yreef], color='orange', ls='--')
angle_current = angle_reef
tmp_angle = np.linspace(0, angle_current, 100)
rtmp = 0.3
plt.plot(0 + rtmp * np.cos(tmp_angle), 0 + rtmp * np.sin(tmp_angle), color='orange', ls='--')
plt.gca().annotate(r'$\theta_{reef}$', xy=(xnew - 0.3, ynew+0.1), xycoords='data', color='orange'

angle_current = angle_old + np.pi
tmp_angle = np.linspace(angle_current, angle_reef, 100)
rtmp = 0.4
plt.plot(0 + rtmp * np.cos(tmp_angle), 0 + rtmp * np.sin(tmp_angle), color='plum', ls='--')
plt.gca().annotate(r'$\theta_{turning}$', xy=(-0.43, -0.34), xycoords='data', color='plum')

off = 0.05
plt.xlim(xreef - off, xold + off)
plt.ylim(yreef - off, yold + off)
plt.gca().set_aspect('equal')
plt.axis('off')
plt.show()
```

```
239.99999999999997
-0.8660254037844384
```



First, the angle of the current trajectory, θ_{actual}, is computed by using the particle position at the previous time step (blue point) and the actual position (red point).

$$\Delta_X = (X_{t-1} - X_t)$$

$$\Delta_Y = (Y_{t-1} - Y_t)$$

$$\theta_{actual} = \arctan 2(\Delta Y, \Delta X) + \pi$$

The direction toward the reef, θ_{reef} is also computed.

$$\Delta_X = (X_{reef} - X_t)$$

$$\Delta_Y = (Y_{reef} - Y_t)$$

$$\theta_{reef} = \arctan 2(\Delta Y, \Delta X)$$

> **Warning**
>
> The angles are computed in the (X, Y) space. Therefore, longitude and latitude coordinates are converted in (X, Y) using the `latlon2xy` Dataset methods.

The turning angle $\theta_{turning}$ is given by:

$$\theta_{turning} = \theta_{reef} - \theta_{actual}$$

The turning angle is then ponderated by the ratio of the distance from the reef to the detection threshold as follows:

$$\theta_{ponderated} = \left(1 - \frac{D}{\beta}\right)\theta_{turning}$$

$$\theta_{ponderated} = \left(1 - \frac{D}{\beta}\right)\left(\theta_{reef} - \theta_{actual}\right)$$

Therefore, the closest to the reef, the strongest the turning angle.

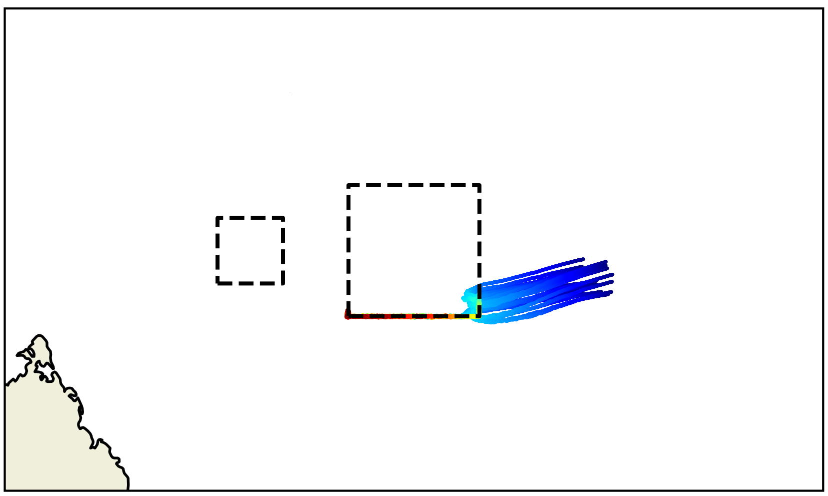
Then, a random angle is picked up following a Von Mises distribution $f(\theta, \theta_{ponderated}, \kappa_{reef})$

An example of a trajectory is provided below. In this case, two target destinations are provided (black boxes). The same κ value was used for both ares (1.2) and the β parameter has been set equal to 3 km.

(ref-orientation-2)=

```python
import xarray as xr
import cartopy.crs as ccrs
import cartopy.feature as cfeature
import os
from glob import glob
import matplotlib.pyplot as plt
import numpy as np
import matplotlib as mpl

config = 'reef'

if os.path.isdir('ichthyop_output'):
    pattern = os.path.join('ichthyop_output', config, '*nc')
else :
    pattern = os.path.join('..', '..', 'ichthyop_output', config, '*nc')
pattern

filelist = glob(pattern)
filelist

data = xr.open_mfdataset(filelist, decode_times=False)
data

# +
lon = data['lon'].values
lat = data['lat'].values
mort = data['mortality'].values

lon = np.ma.masked_where(mort > 0, lon)
lat = np.ma.masked_where(mort > 0, lat)
ntime, ndrifters = lon.shape

# +
time = np.arange(ntime)
drifters = np.arange(ndrifters)

d2d, t2d = np.meshgrid(drifters, time)

# +
plt.figure()

projin = ccrs.PlateCarree()
```

```
projout = ccrs.PlateCarree()

ax = plt.axes(projection = projout)
ax.scatter(lon[:, :], lat[:, :], c=t2d, marker='.', transform=projin, s=0.5, cmap=mpl.colormaps['
feat = ax.add_feature(cfeature.LAND)
feat = ax.add_feature(cfeature.COASTLINE)
xp = np.array([51.625, 52.625, 52.625,51.625, 51.625])
yp = np.array([-1.179878E1, -1.179878E1, -1.079878E1, -1.079878E1, -1.179878E1])
ax.plot(xp, yp, transform=projin, color='k', ls='--')
xp = np.array([51.625, 52.625 - 0.5, 52.625 - 0.5,51.625, 51.625]) - 1
yp = np.array([-1.179878E1 + 0.5, -1.179878E1 + 0.5, -1.079878E1, -1.079878E1, -1.179878E1 + 0.5]
ax.plot(xp, yp, transform=projin, color='k', ls='--')
ax.set_extent([49, 55.25, -13.13, -9.45], crs=projin)
```



7.3.8 Growth

There are 3 different implementations of the growth module. They can be selected by setting the `class_name` parameter in the `action.growth` action configuration block.

7.3.8.1 Linear growth

In the linear growth method described in Lett et al. (2008), length depends on temperature but may also depend on the available food if the `growth.food.enabled` parameter is set to True.

If food dependence is enabled, length increment (in *cm*) is given by:

$$\Delta L = C1 + C2 \times \frac{F}{F + K_S} \times max(T, T_{thres}) \times \Delta t$$

where Δt is the time-step (in seconds), $C1$ and $C2$ are parameters (`coeff1` and `coeff2`), T_{thres} is a temperature threshold (`threshold_temp` parameter), F is the food quantity (read from a biogeochemical dataset), and K_S is a half-saturation constant (`half_saturation` parameter) and T is the temperature (read from the hydrodynamic dataset).

If food dependence is disabled, then increment is provided as follows:

$$\Delta L = C1 + C2 \times max(T, T_{thres}) \times \Delta t$$

The name of the food and temperature variables are provided by the `food_field` and `temperature_field` parameters.

The initial length of the particle is provided by the `initial_length` parameter.

7.3.8.2 Sole growth

In the Sole growth model used in Tanner et al. (2017), length increment (in *cm*) relies on the growth equation from Fonds (1979) :

$$\Delta L = C1 \times T^{C2} \times \Delta t$$

with L the length (in *cm*), T the temperature and Δt the time step seconds, and $C1$ and $C2$ are parameters (`c1` and `c2` respectively).

The temperature field is provided by the `temperature_field` parameter.

7.3.8.3 Exponential growth

In the exponential model, length is given by the following equation:

$$L(t) = L_0 \times \exp(bT^c t)$$

with L_0 the initial length (in cm), b the growth rate and c the temperature dependence of the growth rate.

Taking the derivative of this expression, we obtain the length increment (in *cm*) as a function of time:

$$\Delta L = L \times b \times T^c \times \Delta t$$

The initial length of the particle is provided by the `initial_length` parameter.

7.3.9 Lethal temperature and salinity

In Ichthyop, there is the possibility to define temperature and salinity values below and above which the particle is killed.

The same set of parameters are used in both cases, with the `temperature` and `salinity` prefix. There are two ways to define lethal temperatures and salinities:

- Using constant lethal temperatures or salinities (if `temperature.file.enabled` or `salinity.file.enabled` is set to false)
- Using a CSV file containing lethal temperatures or salinities as a function of the particle class.

If constant lethal temperatures or salinities are provided, the cold and warm lethal temperatures are provided with the `cold_lethal_temperature`, `warm_lethal_temperature`, `fresh_lethal_salinity`, `saline_lethal_salinity`, .

If a CSV file is used, the lethal temperatures and salinities are provided in a CSV (`lethal_temperature_file` and `lethal_salinity_file` parameters) which is formatted as follows:

```
Age(days);Cold temperature (C);Warm temperature (C)
0;14;22
2;13;22
6;12;22
```

The first column provides the class intervals, which are either the age of the particle (in days) or the length of the particle (in cm). The variable that is used to determine the class is provided by the `temperature.class` parameter.

7.3.9.1 DEB growth

The DEB equations are based on Kooijman (2010) and can be found in Flores-Valiente et al. (2023)

Influence of temperature is given by equation 5 of Flores-Valiente et al. (2023):

$$c_T = \exp\left(\frac{T_A}{T_1} - \frac{T_A}{T}\right)\left[\frac{1 + \exp\left(\frac{T_{AL}}{T_1} - \frac{T_{AL}}{T_L}\right) + \exp\left(\frac{T_{AH}}{T_H} - \frac{T_{AH}}{T_1}\right)}{1 + \exp\left(\frac{T_{AL}}{T} - \frac{T_{AL}}{T_L}\right) + \exp\left(\frac{T_{AH}}{T_H} - \frac{T_{AH}}{T}\right)}\right] \tag{7.3}$$

If the particle is an embryo ($E_H < E_{Hb}$), then it does not feed. Please note that, according to this definition, a larva that hatched but does not feed is considered an embryo in DEB theory. Else, the scaled functional response f is computed as:

$$f = \frac{F}{F + K} \tag{7.4}$$

with F the food vector and K the half-saturation constant.

Assimilation rates $\{\dot{p_{Am}}\}$, energy conductance $\dot{v}$, volume specific somatic maintenance ($[\dot{p}_M]$) and maturity maintenances rates ($\dot{k}_J$) are corrected by the c_T factor.

If the accelerated DEB model is used (see e.g. Kooijman et al. (2011) and Kooijman (2014) for more details), the acceleration factor is computed as follows

$$s_M = \begin{cases} 1 & (E < E_{Hb}) \\ \dfrac{V^{1/3}}{L_b} & (E_{Hb} \leq E_H < E_{Hj}) \\ \dfrac{L_j}{L_b} & (E_H > E_{Hj}) \end{cases} \tag{7.5}$$

with L_b the structural length of the particle when $E_H = E_H b$ (i.e the length at birth, i.e. first feeding) and L_j the structural length of the particle when $E_H = E_H j$ (i.e the length when metamorphosis occurs). The s_M factor is then used to multiply the assimilation rate $\{\dot{p_{Am}}\}$ and the energy conductance $\dot{v}$.

(NICOLAS: s'agit il de la longueur structurelle $L = V^{1/3}$ ou de la longueur mesurée $L_w = L/\delta_M$). Please note that L_b and L_j are here considered as parameters but they should be in fact calculated prior any Ichthyop runs at (e.g. at $f = 1$) as they depend on the other parameters. If you are considering testing different parameter sets, L_b and L_j should be modified accordingly (e.g. using the DEBtool function `get_tj.m`).

Consequently, the assimilation (in $J.d^{-1}$) is computed as follows:

$$\dot{p}_A = \begin{cases} 0 & (E_H < E_{Hb}) \\ c_T s_M f \{\dot{p}_{Am}\} V^{2/3} & (E_H \geq E_{Hb}) \end{cases} \tag{7.6}$$

Energy loss to maintenance is given by

$$\dot{p}_M = c_T [\dot{p}_M] V \tag{7.7}$$

The mobilized energy is given by:

$$\dot{p}_C = c_T \frac{\frac{E}{V}\left([E_G] s_M \dot{v} V^{2/3} + \dot{p}_M\right)}{\kappa \frac{E}{V} + [E_G]} \tag{7.8}$$

Maturity maintenance is given by

$$\dot{p}_J = c_T \dot{k}_J E_H \tag{7.9}$$

Energy that is sent to structural growth is given by:

$$\dot{p}_G = \kappa \dot{p}_C - \dot{p}_M \tag{7.10}$$

Energy that goes to development (as embryo and juvenile) or to reproduction (as adults) is given by:

$$\dot{p}_R = (1-\kappa)\dot{p}_C - \dot{p}_J \tag{7.11}$$

Finally, the increment of reserve E and structural volume V is given by:

$$\frac{dE}{dt} = \dot{p}_A - \dot{p}_C \tag{7.12}$$

$$\frac{dV}{dt} = \frac{\dot{p}_G}{[E_G]} \tag{7.13}$$

The increment of the energy invested into development or reproduction depends on the value of E_H.

$$\frac{dE_H}{dt} = \begin{cases} \dot{p}_R & E_H < E_{Hp} \\ 0 & \text{otherwise} \end{cases} \tag{7.14}$$

$$\frac{dE_R}{dt} = \begin{cases} 0 & E_H < E_{Hp} \\ \dot{p}_R & \text{otherwise} \end{cases} \tag{7.15}$$

Finally, starvation mortality is determined whether one of the following conditions are matched:

$$M_{starv} = \begin{cases} \kappa \dot{p}_C < \dot{p}_M \\ (1-\kappa)\dot{p}_C < \dot{p}_J \end{cases} \tag{7.16}$$

Finally, all the states variables are then incremented:

$$E(t + \Delta t) = E(t) + \frac{dE}{dt}\Delta t \tag{7.17}$$

$$V(t + \Delta t) = V(t) + \frac{dV}{dt}\Delta t \tag{7.18}$$

$$E_H(t + \Delta t) = E_H(t) + \frac{dE_H}{dt}\Delta t \tag{7.19}$$

$$E_R(t + \Delta t) = E_R(t) + \frac{dE_R}{dt}\Delta t \tag{7.20}$$

Additional references

Kooijman, S. A. L. M. (2010). Dynamic Energy Budget theory for metabolic organisation. Cambridge Univ. Press, Cambridge. doi: 10.1098/rstb.2010. Kooijman, S. A. L. M. (2014). Metabolic acceleration in animal ontogeny: An evolutionary perspective. Journal of Sea Research, 94(Supplement C), 128–137. https://doi.org/10.1016/j.seares.2014.06.005 Kooijman, S. A. L. M., Pecquerie, L., Augustine, S., & Jusup, M. (2011). Scenarios for acceleration in fish development and the role of metamorphosis. J. Sea Res., 66, 419–423. https://doi.org/10.1016/j.seares.2011.04.016

(**book?**){kooijman_dynamic_2010, title = {Dynamic {Energy} {Budget} theory for metabolic organisation}, isbn = {978-0-521-13191-9}, url = {doi: 10.1098/rstb.2010.}, publisher = {Cambridge Univ. Press, Cambridge}, author = {Kooijman, S.A.L.M.}, year = {2010}, }

(**article?**){kooijman_scenarios_2011, title = {Scenarios for acceleration in fish development and the role of metamorphosis.}, volume = {66}, doi = {https://doi.org/10.1016/j.seares.2011.04.016}, journal = {J. Sea Res.}, author = {Kooijman, S. A. L. M. and Pecquerie, L. and Augustine, S. and Jusup, M.}, year = {2011}, pages = {419–423}, }

(**article?**){kooijman_metabolic_2014, series = {Dynamic {Energy} {Budget} theory: applications in marine sciences and fishery biology}, title = {Metabolic acceleration in animal ontogeny: {An} evolutionary perspective}, volume = {94}, issn = {1385-1101}, shorttitle = {Metabolic acceleration in animal ontogeny}, url = {http://www.sciencedirect.com/science/article/pii/S1385110114001105}, doi = {10.1016/j.seares.2014.06.005}, number = {Supplement C}, urldate = {2017-11-21}, journal = {Journal of Sea Research}, author = {Kooijman, S. A. L. M.}, month = nov, year = {2014}, keywords = {Acceleration, Dispersal, Maturation, Parental care, add_my_pet}, pages = {128–137}, }

Part III

Developer documentation

In this section, the Ichthyop console is described.

8 Java install

Beforehand, let us clarify some of the acronyms regarding the Java technologies.

`JVM`: Java Virtual Machine. It is a set of software programs that interprets the Java byte code.

`JRE`: Java Runtime Environment. It is a kit distributed by Sun to execute Java programs. A `JRE` provides a `JVM` and some basic Java libraries.

`JDK` or `SDK`: Java (or Software) Development Kit bound to the programmer. It provides a `JRE`, a compiler, useful programs, examples and the source of the API (Application Programming Interface: some standard libraries).

It is strongly recommended to download a `JDK`, in order to both compile and run the model. Builds for different platforms can be found here.

9 Manager initialization

When an Ichthyop simulation is launched, managers are mobilized in the following order (cf. `SimulationManager.mobiliseManagers()` method):

This order is the one in which the managers will be setup and then initialized.

During the setup process, the `setupPerformed` methods, implemented on all the manager classes, will be called.

After this setup process, the managers will be initialized. This will be achieved by calling the `initializePerformed` methods, implemented on all the manager classes.

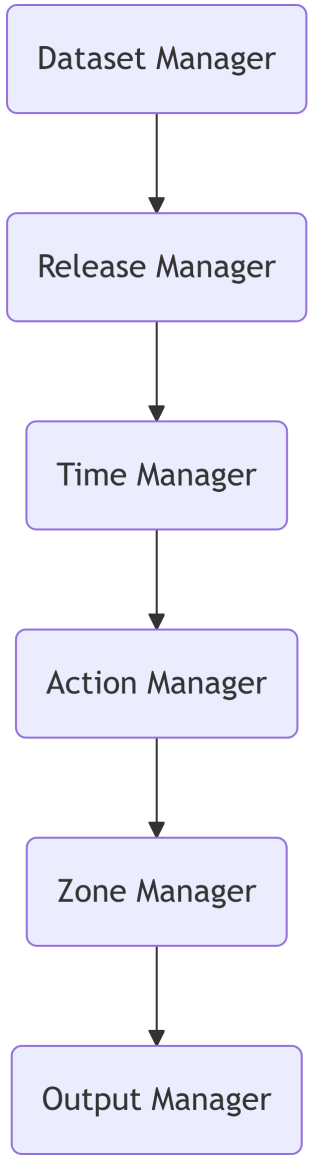


Figure 9.1: Order in which managers are mobilized.

10 Particles

Ichthyop particles (`Particle.java` class) contains the attributes described in Table 10.1.

Table 10.1: Particle state variables

Variable	Description	Type	Units
x	Particle x position within the grid	float, $\in [0, N_x - 1]$	
y	Particle y position within the grid	float, $\in [0, N_y - 1]$	
z	Particle z position within the grid	float, $\in [0, N_z - 1]$	
dx	Particle increment/decrement of x position	float	
dy	Particle increment/decrement of y position	float	
dz	Particle increment/decrement of z position	float	
lat	Particle longitude	float	Degrees East
lon	Particle latitude	float	Degrees North
depth	Particle depth	float, < 0	m
index	Particle index	int, $\in [0, N_{part} - 1]$	
age	Particle age	int	seconds
deathCause	Mortality status	`ParticleMortality`	
living	True if a particle is alive	bool	
locked	True if a particle is locked	bool	
layers	List of additional particle layers	`List<ParticleLayer>`	

When using Ichthyop with additional processes, such as growth or DEB processes, additional variables need to be tracked.

This is done by firsr creating a layer class, which should extend the `ParticleLayer.java` class. An exemple if provided as follows:

```
package org.previmer.ichthyop.particle;

public class LengthParticleLayer extends ParticleLayer {
```

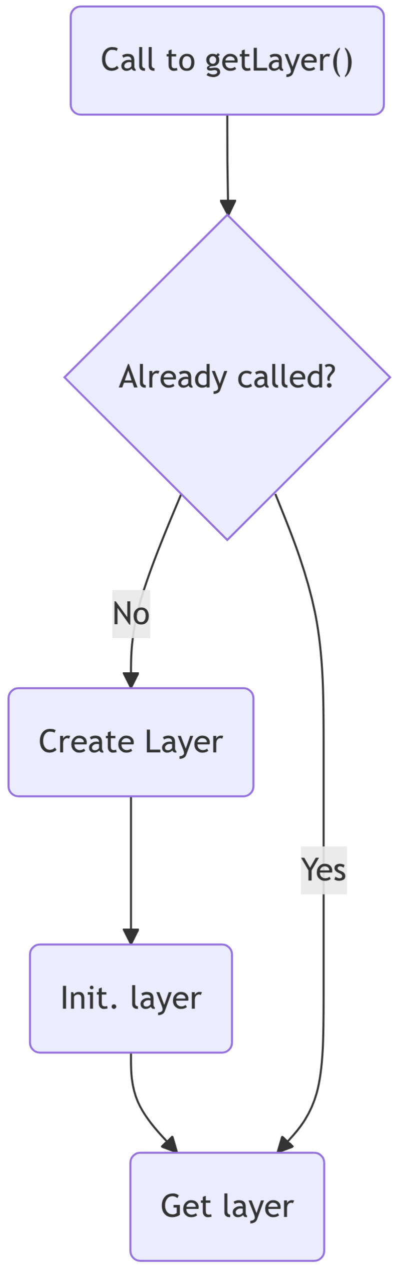
```java
    private double length;

    public LengthParticleLayer(IParticle particle) {
        super(particle);
    }

    @Override
    public void init() {
        length = 0;
    }

    public double getLength() {
        return length;
    }

    public void incrementLength(double dlength) {
        length += dlength;
    }
}
```

This class contains additional particle attributes (here, for instance `length`) and methods that can return and modify these attributes.

These new variables can be accessed as follows:

```java
LengthParticleLayer lengthLayer = (LengthParticleLayer) particle.getLayer(LengthParticleLayer.cla
lengthLayer.setLength(length_init);
```

During the first call to the `getLayer` method applied to a given layer class, the latter will be instanciated, initialized and add to the particle's `layers` list attribute, as shown below.



Order in which managers are mobilized.

11 Grid management

11.0.1 NEMO grid

In this section, the main features of the NEMO grid and the implications in Ichthyop are summarized.

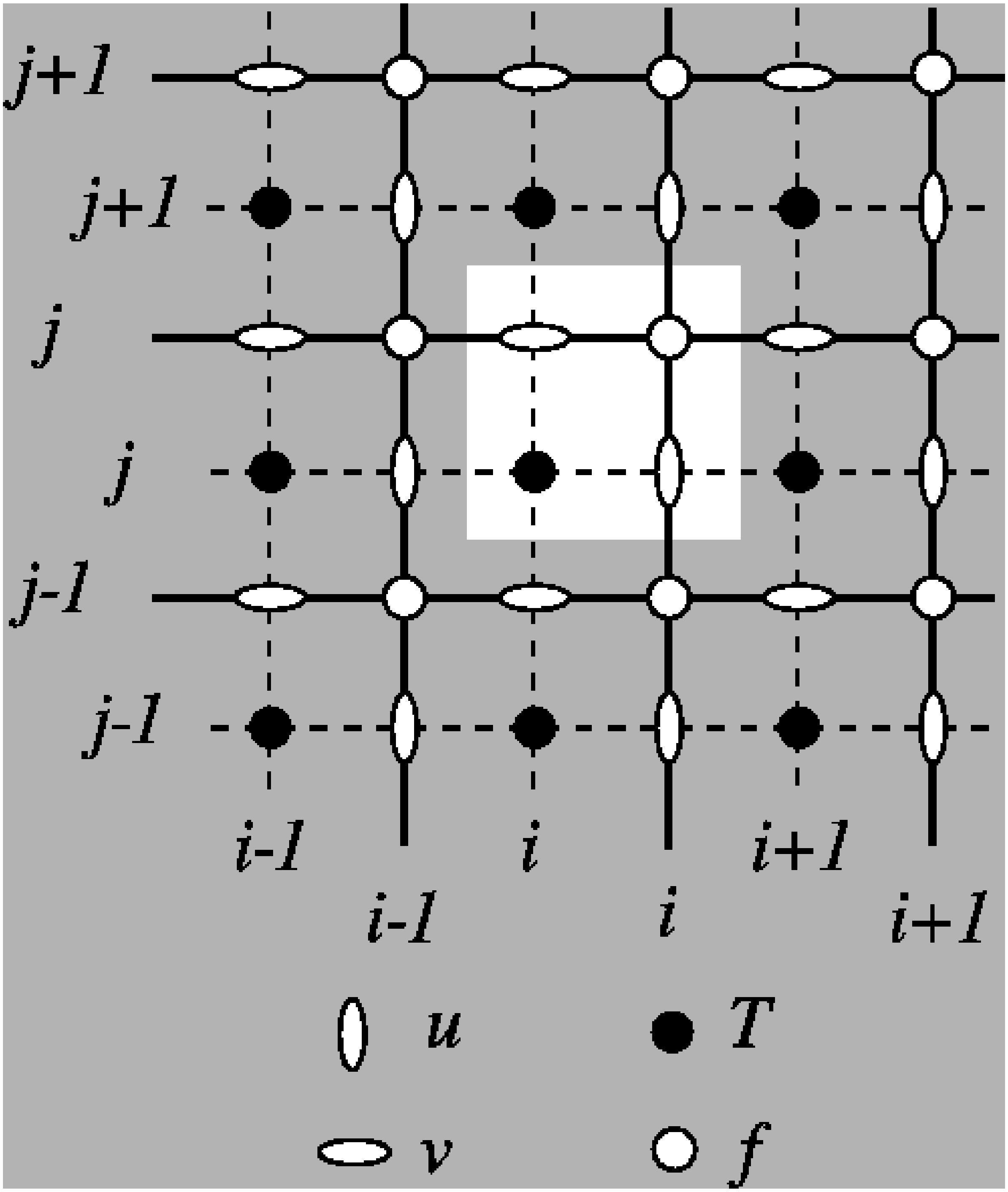
11.0.1.1 Horizontal

11.0.1.1.0.1 Computation domain

In the NEMO grid, the computation domain extends from the western edge of the western T cell (`i=0`) to the eastern edge of the eastern T cells (`i=nx`), and from the southern edge of the southern T cells (`j=0`) to the northern edge of the nothern T cells (`j=ny`).

11.0.1.1.0.2 Indexing

The horizontal layout of the NEMO grid is as follows:



Tracer points (T points) are stored at the center of the cell. Zonal velocities (U points) are stored on the eastern face, while meridional velocities (V points) are stored on the northern face. **U, V and T points have the same number of elements!**

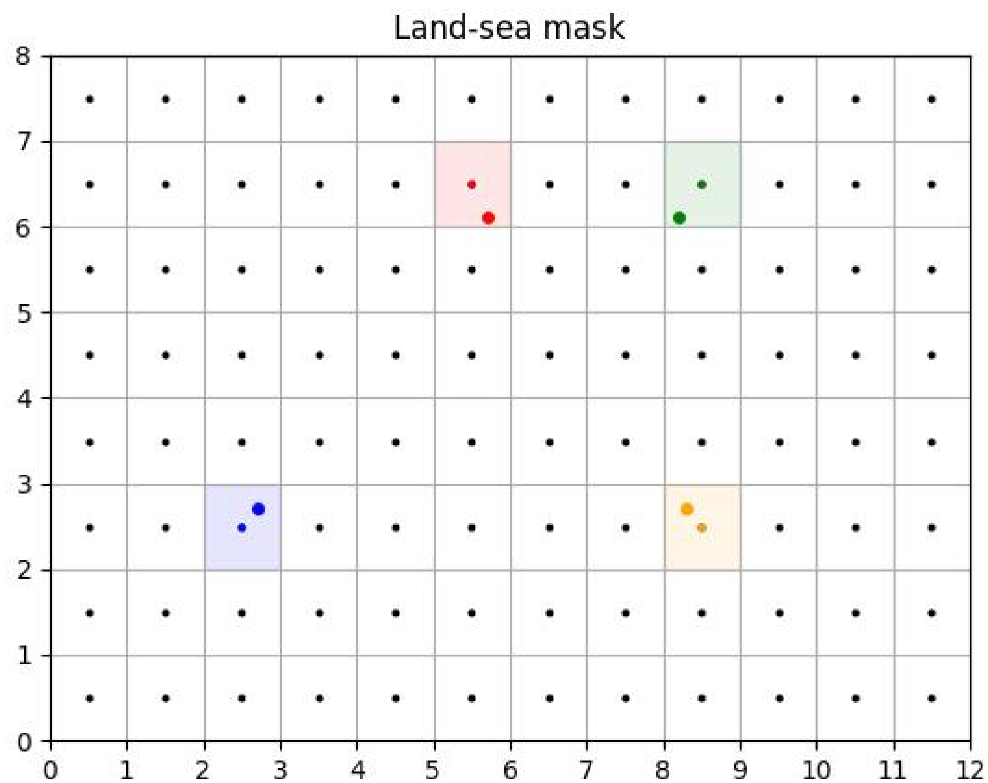
11.0.1.1.0.3 Scale factors

In NEMO, the zonal and meridional length of the cells are stored in the `e1x` and `e2x` variables, with `x` equals to `t`, `u` or `v` depending on the point considered.

11.0.1.1.0.4 Land-sea mask

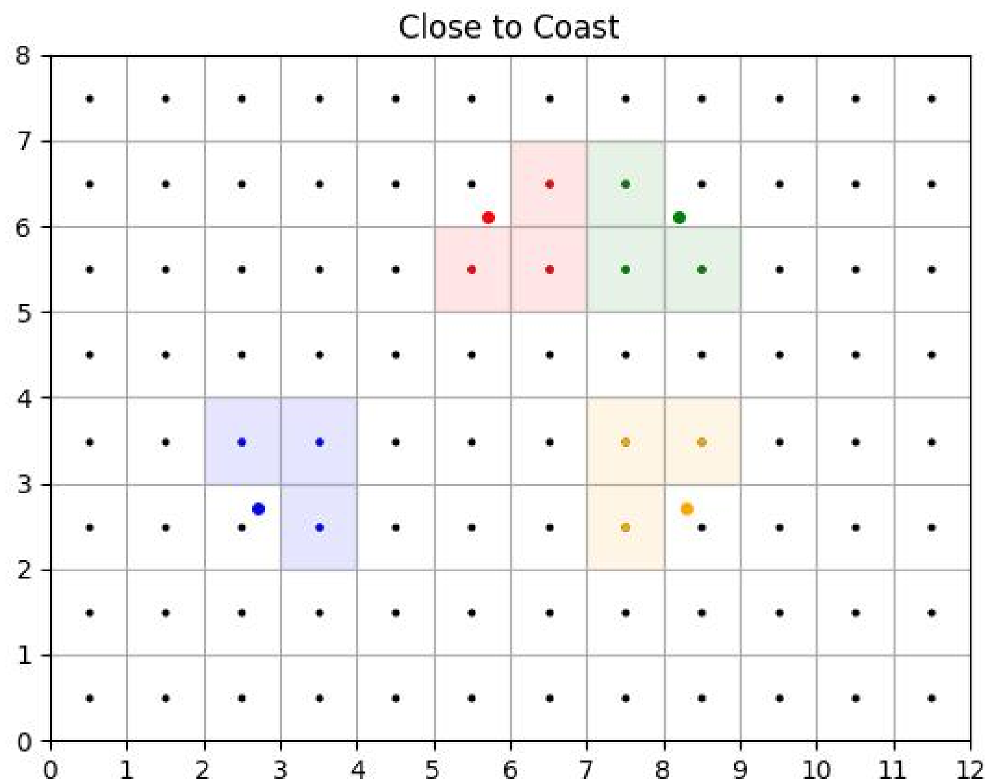
For a particle at a given location, we determine whether it is on land as follows:

- We extract the `i` index of the `T` land-sea mask to use by computing `round(x - 0.5)`.
- We extract the `j` index of the `T` land-sea mask to use by computing `round(y - 0.5)`.
- We extract the mask value at the `(i, j)` location.



11.0.1.1.0.5 Close to coast

To determine whether a particle is close to coast, we extract the three neighbouring T cells. If one of them is land, then it assumed to be close to coast.



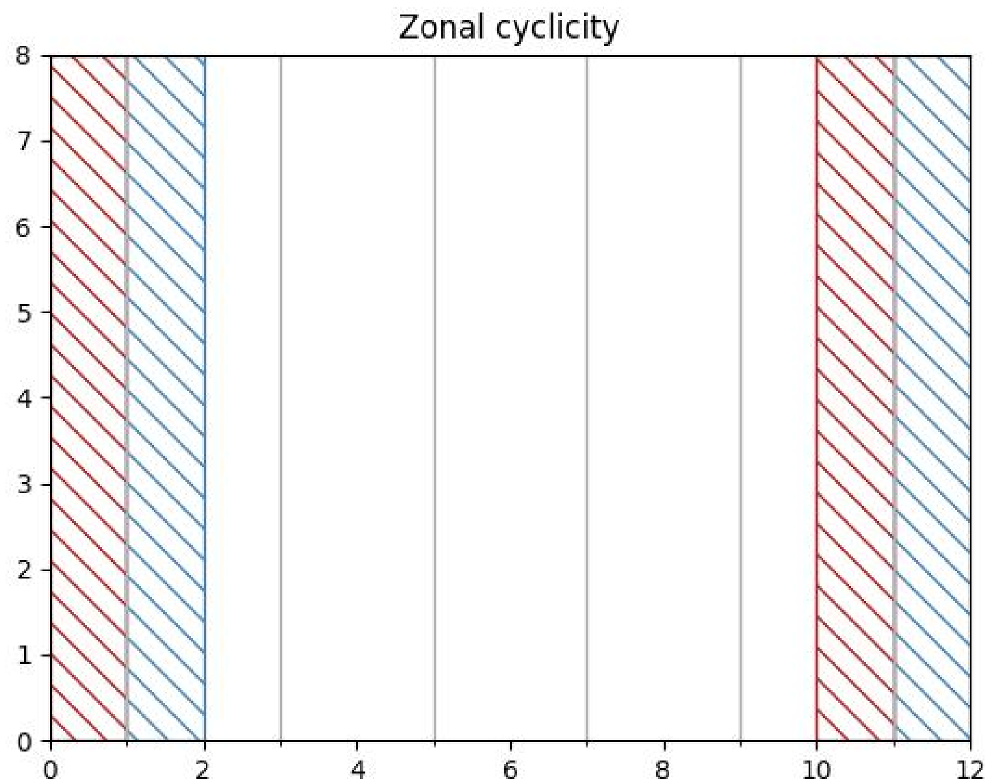
11.0.1.1.0.6 Is On Edge

The particle is considered to be out of the domain if the `y` value is greater than `ny - 0.5` (no possible interpolation of `T` points) or less than `1` (no possible interpolation of `V` points).

If there is no zonal cyclicity, the particle is also considered to be out of the domain if the `x` value is greater than `nx - 0.5` (no possible interpolation of `T` points) or less than `1` (no possible interpolation of `U` points).

11.0.1.1.0.7 Zonal cyclicity

For regional simulations, there is no zonal cyclicty. On the other hand, for global NEMO simulations, which runs on the ORCA grid, the zonal cyclicity is as follows (indexes are provided for T points):



Therefore:

- if $x \leq 1$, the particle is moved at $N_x - 2 + x$
- if $x \geq nx - 1$, the particle is moved at $x - N_x + 2$)

11.0.1.1.0.8 Interpolation

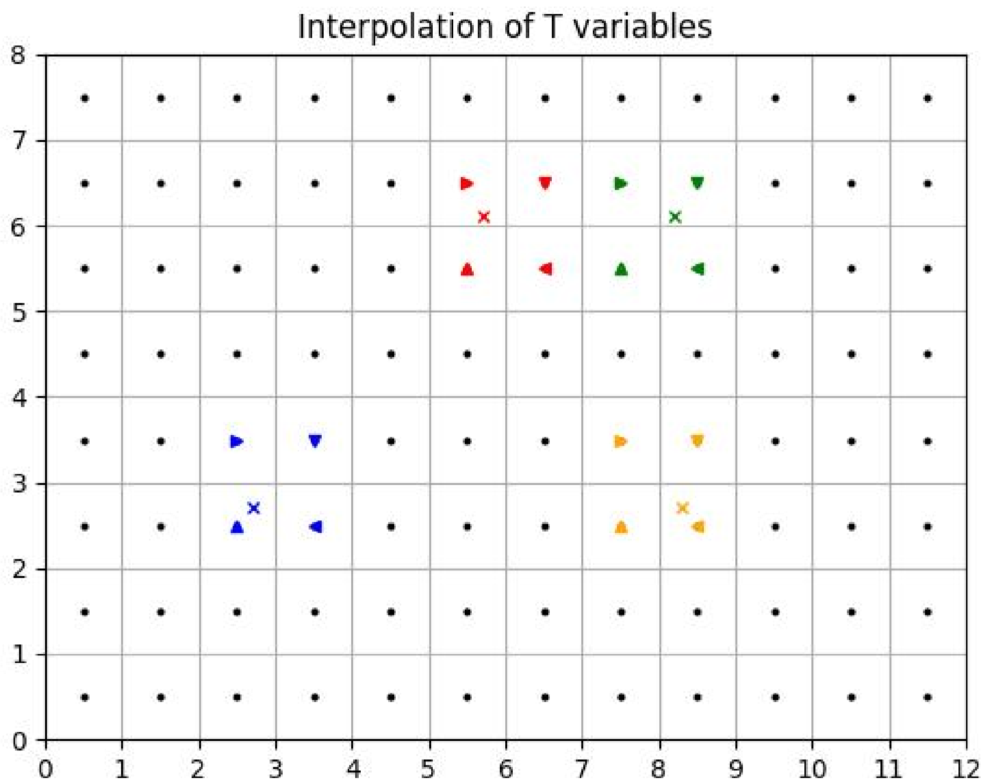
Interpolation of T variables

Given a given position index of a particle with the `T` grid, the determination of the interpolation is done as follows:

- First, the `i` index of the `T` grid column left of the particle is found. This is done by using `floor` on the `x - 0.5` value. The removing of 0.5 is to convert the `x` value from the computational grid to the `T` grid.
- Then, the `j` index of the `T` grid line below the particle is found. This is done by using `floor` on the `y - 0.5` value. The removing of 0.5 is to convert the `y` value from the computational grid to the `T` grid.

- The area to consider is defined by the `[i, i + 1]` and `[j, j + 1]` squares.

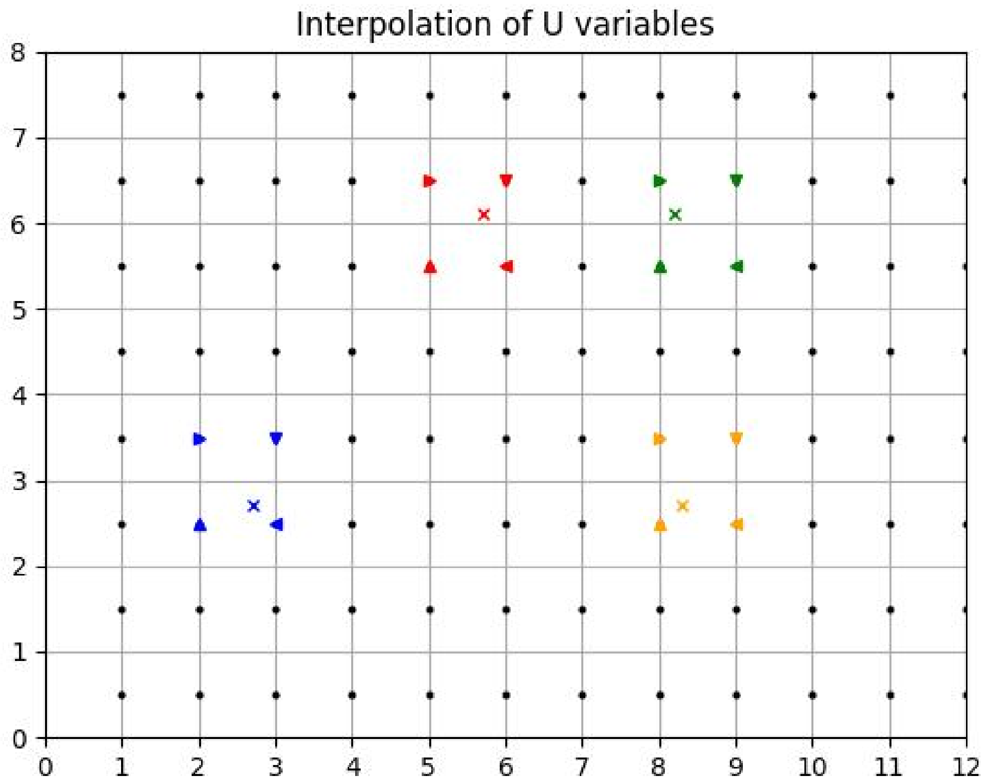
An illustration is given below



Interpolation of U variables

Interpolation of `U` variables is done as follows:

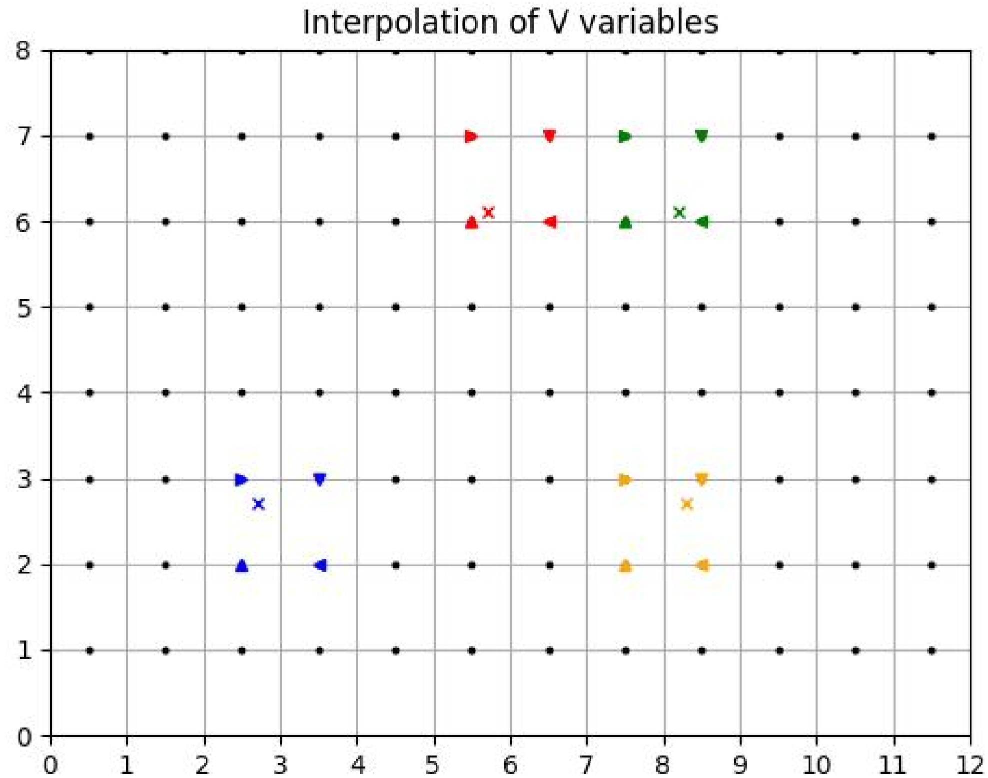
- First, the `i` index of the `U` point left of the particle is found by using `floor(x - 1)`. The `-1` is to move from the computation grid to the `U` grid system.
- Then, the `j` index of the `U` grid line below the particle is found. This is done by using `floor` on the `y - 0.5` value. The `-0.5` is to move from the computation grid to the `U` grid system.
- The box used to average the variable is therefore defined by the `[i, i + 1]` and `[j, j + 1]` squares.



Interpolation of V variables

Interpolation of `V` variables is done as follows:

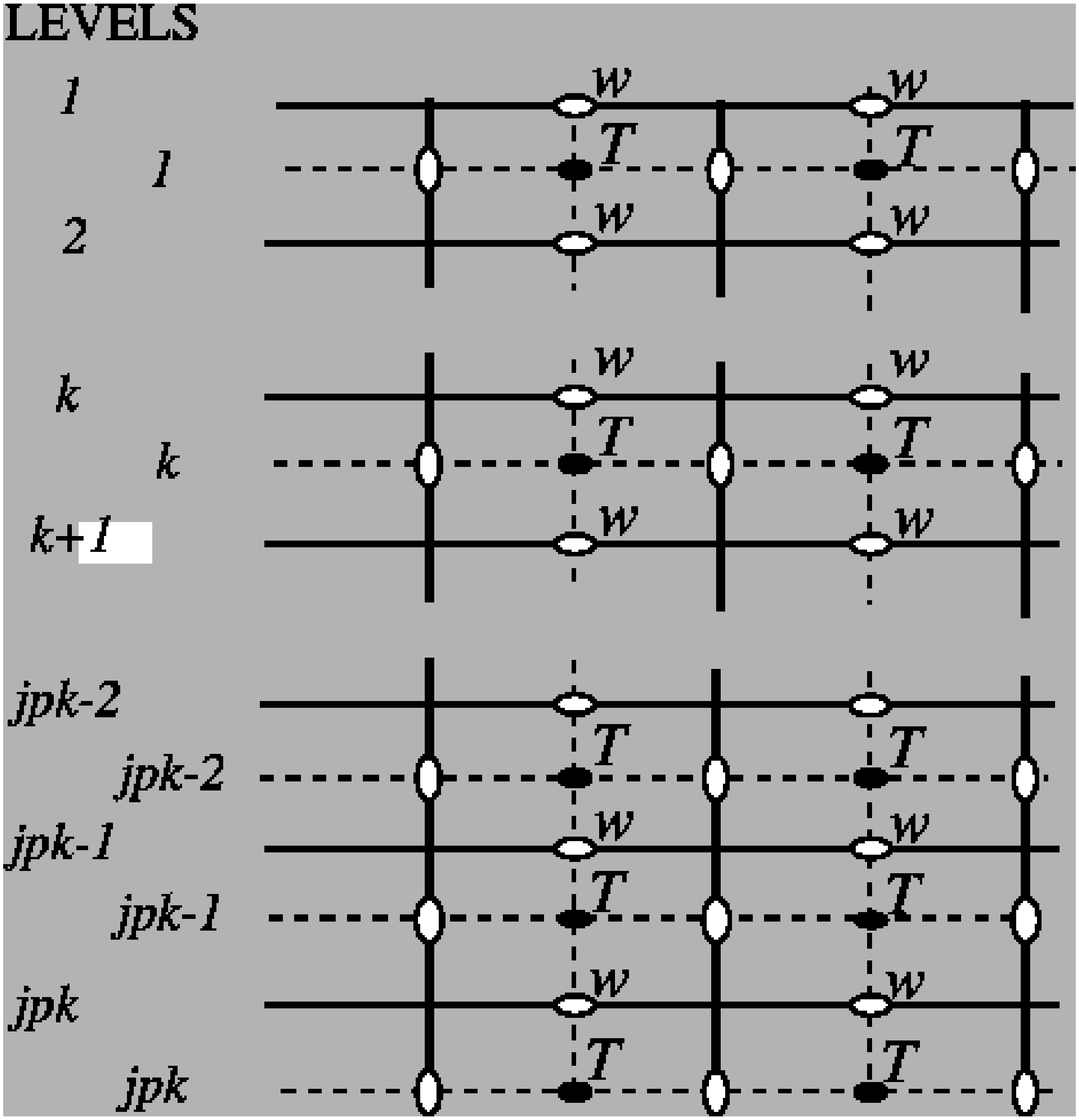
- First, the `i` index of the `V` point left of the particle is found by using `floor(x - 0.5)`. The `-0.5` is to move from the computation grid to the `U` grid system.
- Then, the `j` index of the `V` grid line below the particle is found. This is done by using `floor` on the `y - 1` value. The `-1` is to move from the computation grid to the `U` grid system.
- The box used to average the variable is therefore defined by the `[i, i + 1]` and `[j, j + 1]` squares.



11.0.1.2 Vertical

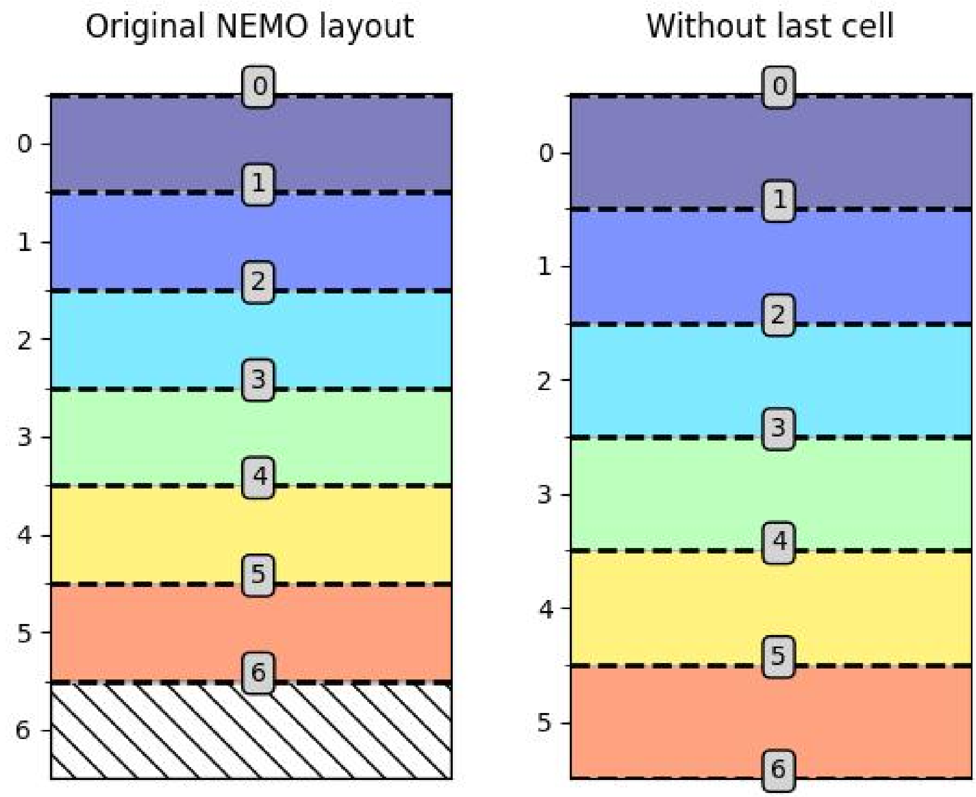
11.0.1.2.0.1 Indexing

The original NEMO vertical indexing system is shown below:



Index starts at 0 (at the surface) and ends at $N_z - 1$ at depth. There are as many `W` levels as `T` levels. In NEMO, the `T` point situated at $N_z - 1$ **is always masked**. `W` levels are located *above* the corresponding `T` points.

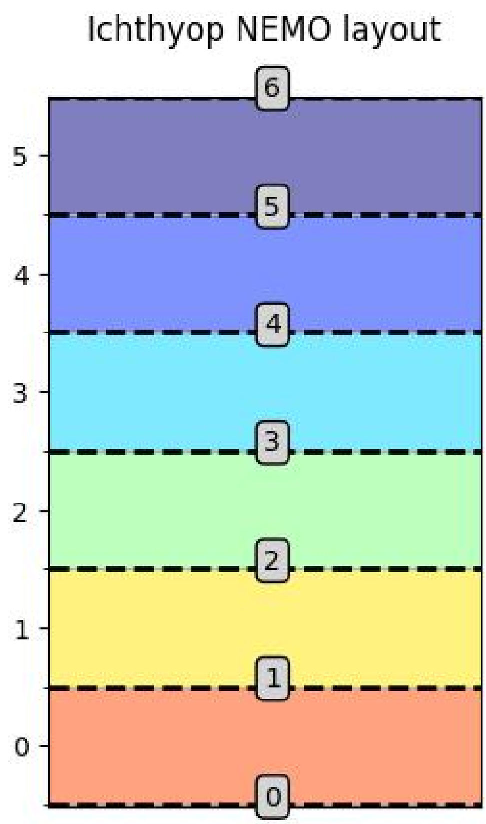
Therefore, in Ichthyop, only the first $N_z - 1$ `T` points are read, while all the N_z `W` points are read. This is show below:



Vertical indexing system as used in Ichthyop. Red dashed lines represent the `W` levels (cell edges), whose index is given in the gray box.

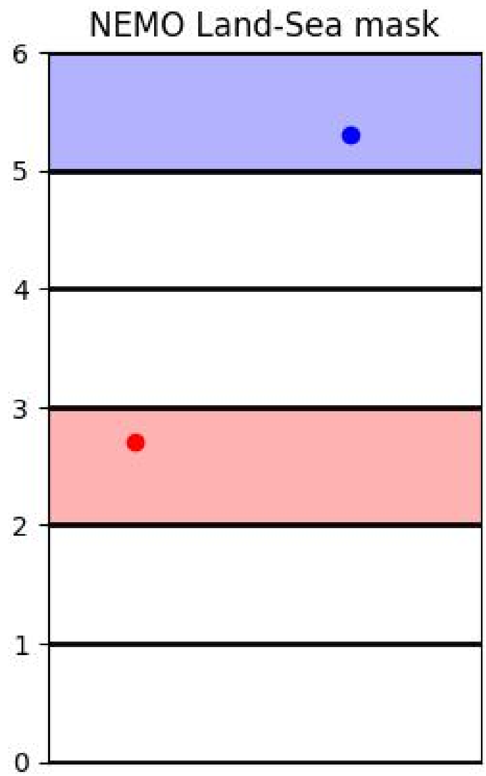
There is now N_z W levels but $N_z - 1$ T levels.

Furthermore, since in Ichthyop the first index corresponds to the seabed, the arrays are vertically flipped. Consequently, the final structure of the vertical grid is as follows:



Corrected vertical indexing, with `k=0` associated with the bottom depth.

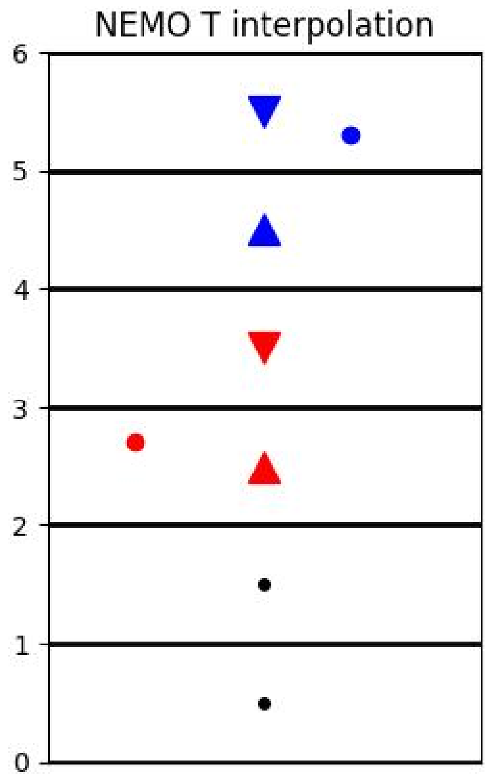
11.0.1.2.0.2 Land-sea mask



Vertical land-sea mask

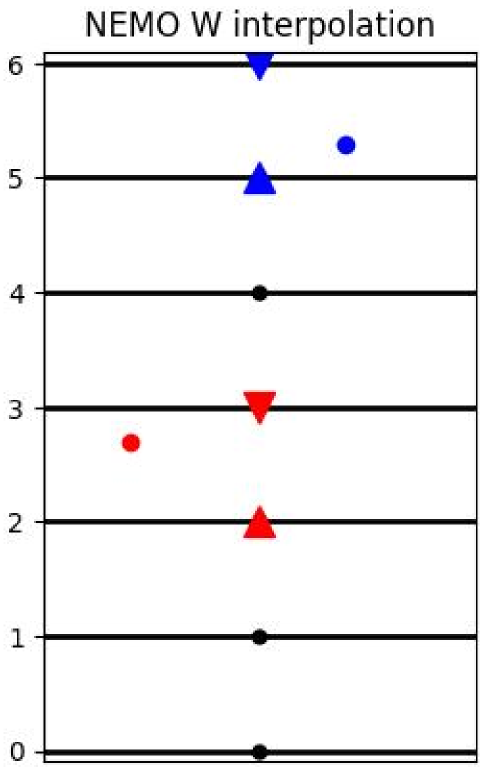
11.0.1.2.0.3 Interpolation

T variables



Vertical interpolation of T variables.

W variables



:align: center :width: 250 px

Vertical interpolation of T variables.

11.0.2 ROMS grid

In this section, the main features of the ROMS grid and the implications in Ichthyop are summarized.

11.0.2.1 Horizontal

The horizontal grid layout of the ROMS model is shown below:

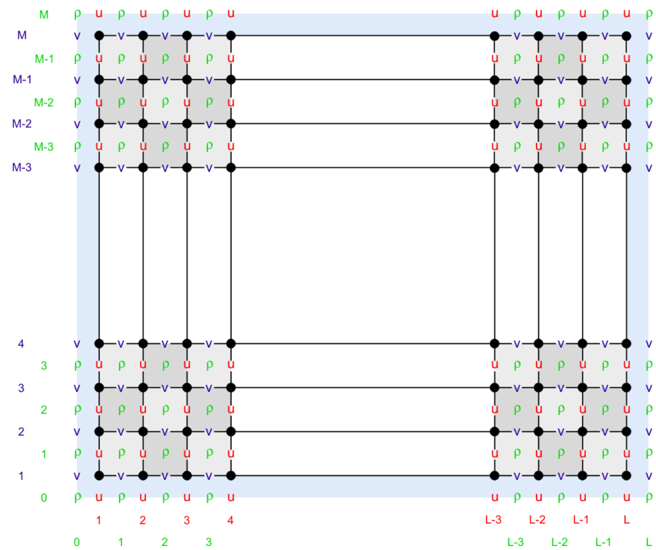


Figure 11.1: ROMS staggered grid structure

Contrary to NEMO and MARS, the `U` points are located on the *western* face, while velocities are located on the *southern* faces. However, However, as indicated on the Wikiroms page, while the `T` interior domain has (N_y, N_x) dimensions, the `U` domain is (N_y, N_x-1) points while and `V` domain is (N_y-1, N_x) points. Indeed, the first row for V and the first column for U are discarded. Therefore, the structure of the input ROMS grid is as follows:

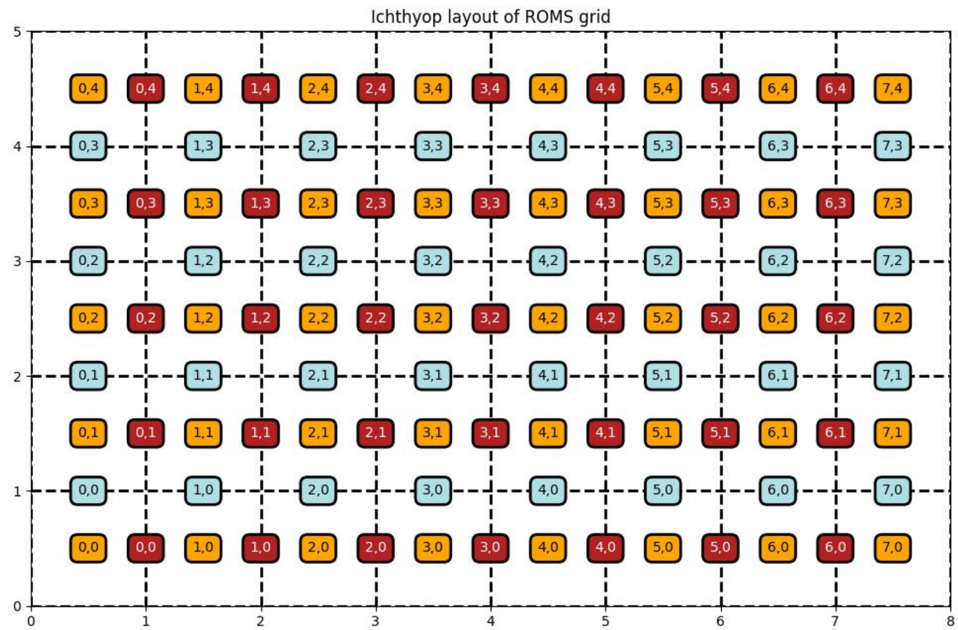


Figure 11.2: ROMS grid as interpreted by Ichthyop

11.0.2.1.1 Land-sea mask

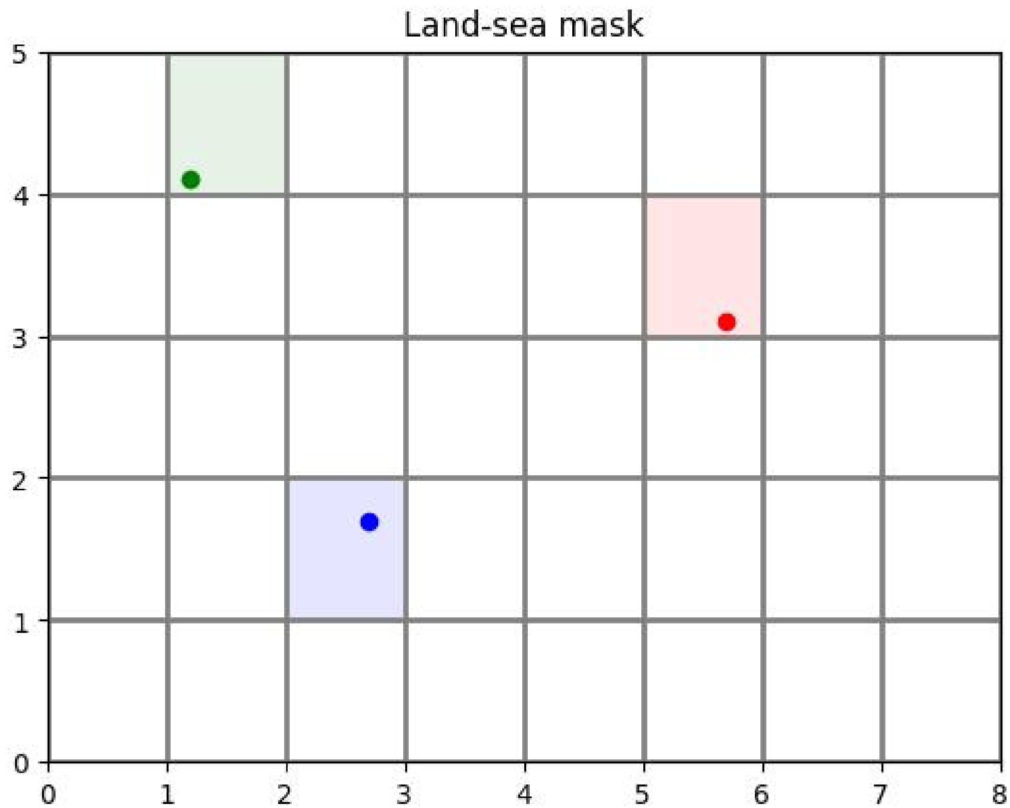


Figure 11.3: Land-sea masking for ROMS grid

11.0.2.1.2 Interpolation

11.0.2.1.2.1 T interpolation

Given a given position index of a particle with the `T` grid, the determination of the interpolation is done as follows:

- First, the `i` index of the `T` grid column left of the particle is found. This is done by using `floor` on the `x - 0.5` value. The removing of 0.5 is to convert the `x` value from the computational grid to the `T` grid.
- Then, the `j` index of the `T` grid line below the particle is found. This is done by using `floor` on the `y - 0.5` value. The removing of 0.5 is to convert the `y` value from the computational grid to the `T` grid.
- The area to consider is defined by the `[i, i + 1]` and `[j, j + 1]` squares.

An illustration is given below

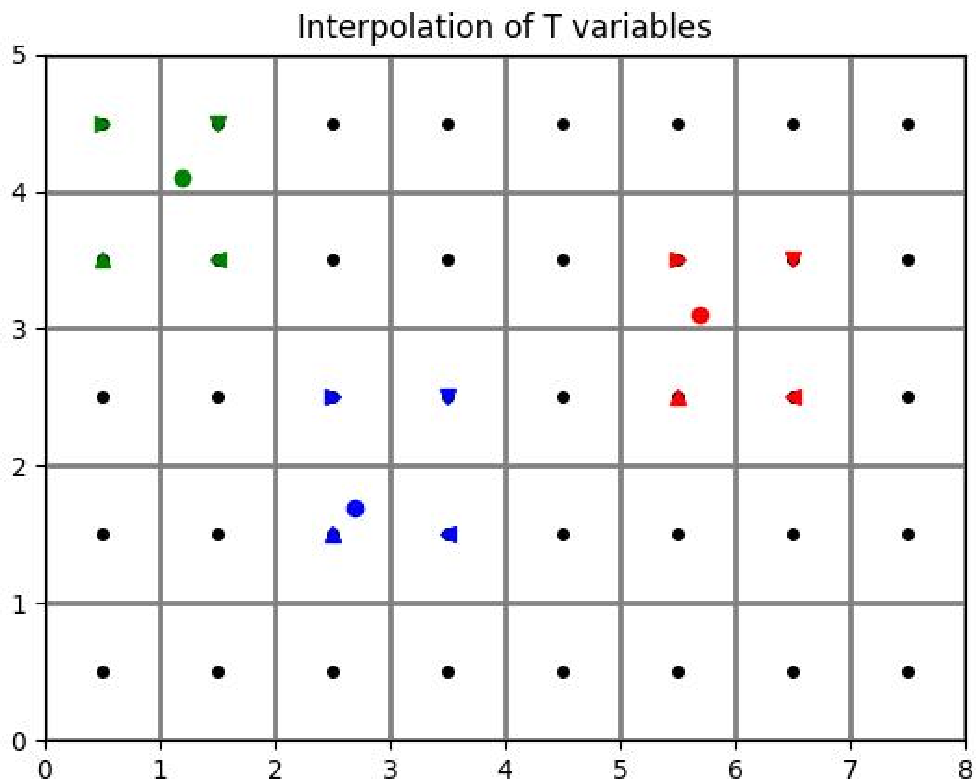


Figure 11.4: Interpolation of T points from ROMS grid

11.0.2.1.2.2 U interpolation

Interpolation of `U` variables is done as follows:

- First, the `i` index of the `U` point left of the particle is found by using `floor(x - 1)`. The `-1` is to move from the computation grid to the `U` grid system.
- Then, the `j` index of the `U` grid line below the particle is found. This is done by using `floor` on the `y - 0.5` value. The `-0.5` is to move from the computation grid to the `U` grid system.
- The box used to average the variable is therefore defined by the `[i, i + 1]` and `[j, j + 1]` squares.

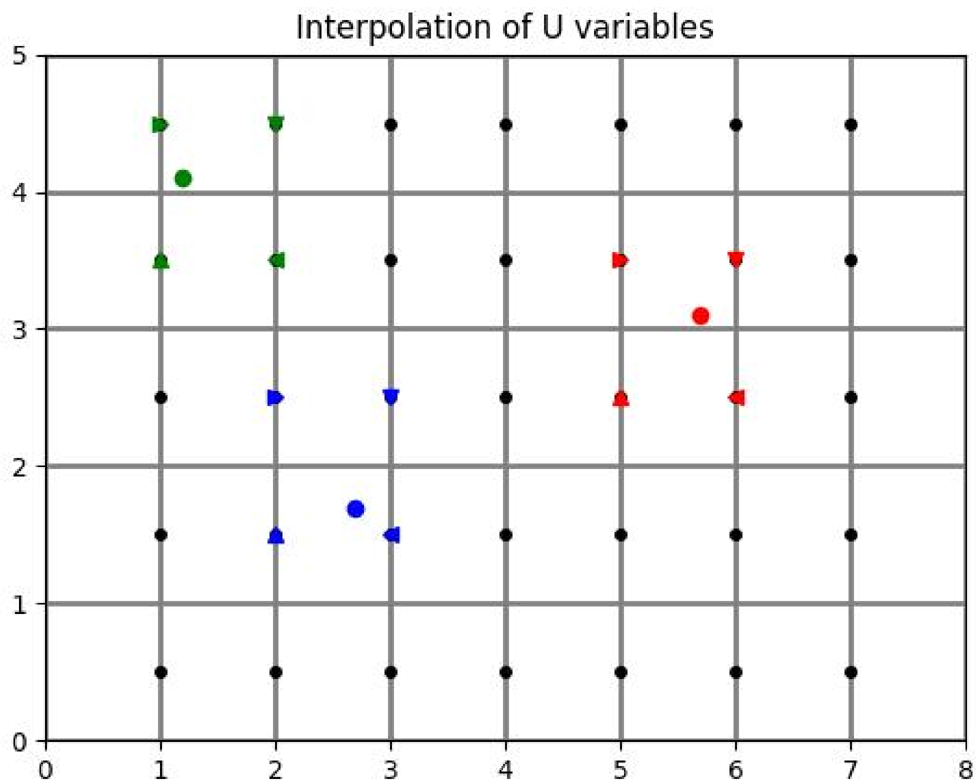


Figure 11.5: Interpolation of U points from ROMS grid

11.0.2.1.2.3 V interpolation

Interpolation of `V` variables is done as follows:

- First, the `i` index of the `V` point left of the particle is found by using `floor(x - 0.5)`. The `-0.5` is to move from the computation grid to the `U` grid system.
- Then, the `j` index of the `V` grid line below the particle is found. This is done by using `floor` on the `y - 1` value. The `-1` is to move from the computation grid to the `U` grid system.
- The box used to average the variable is therefore defined by the `[i, i + 1]` and `[j, j + 1]` squares.

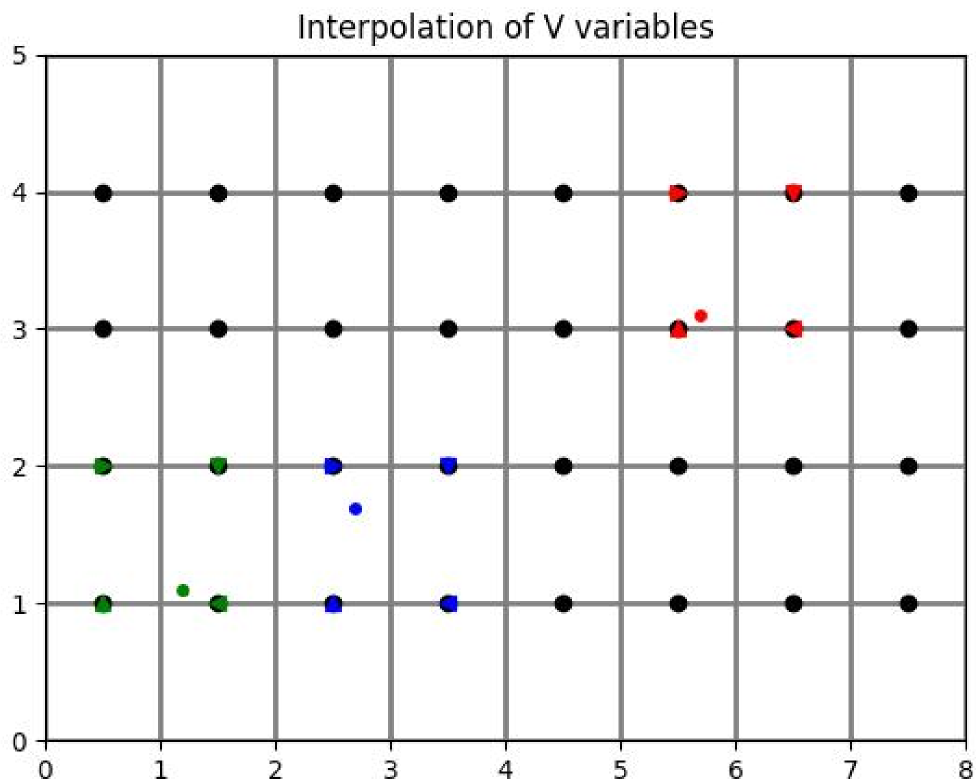


Figure 11.6: Interpolation of V points from ROMS grid

11.0.2.1.3 Is on edge

A particle is considered to be out of domain when $x \leq 1$ (no possible interpolation of U on the western face), when $y \geq N_x - 1$ (no possible interpolation of U on the eastern face), when $y \leq 1$ (no possible interpolation of V on the southern domain) or when $y \geq N_y - 1$ (no possible interpolation of V on the northern part of the domain).

The excluded domain is represented below:

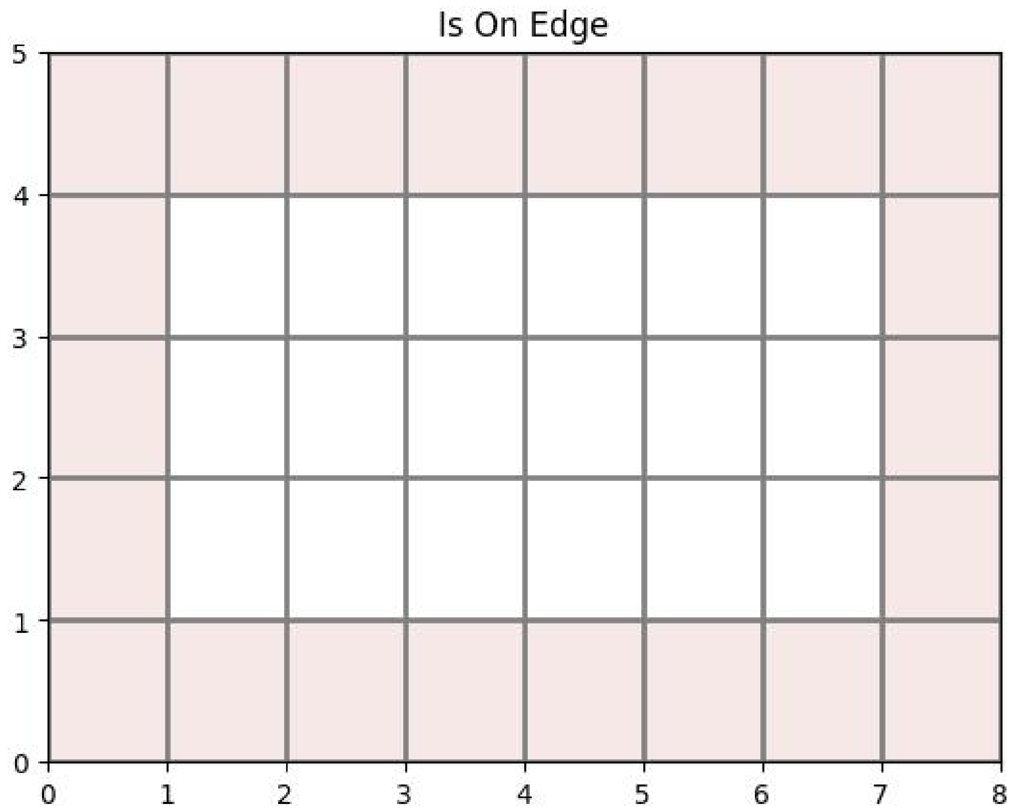


Figure 11.7: Excluded domain in the Ichthyop ROMS simulations.

11.0.2.2 Vertical

11.0.2.2.1 Sigma coordinate

The vertical coordinate system of ROMS is discussed on WikiRoms and shown below.

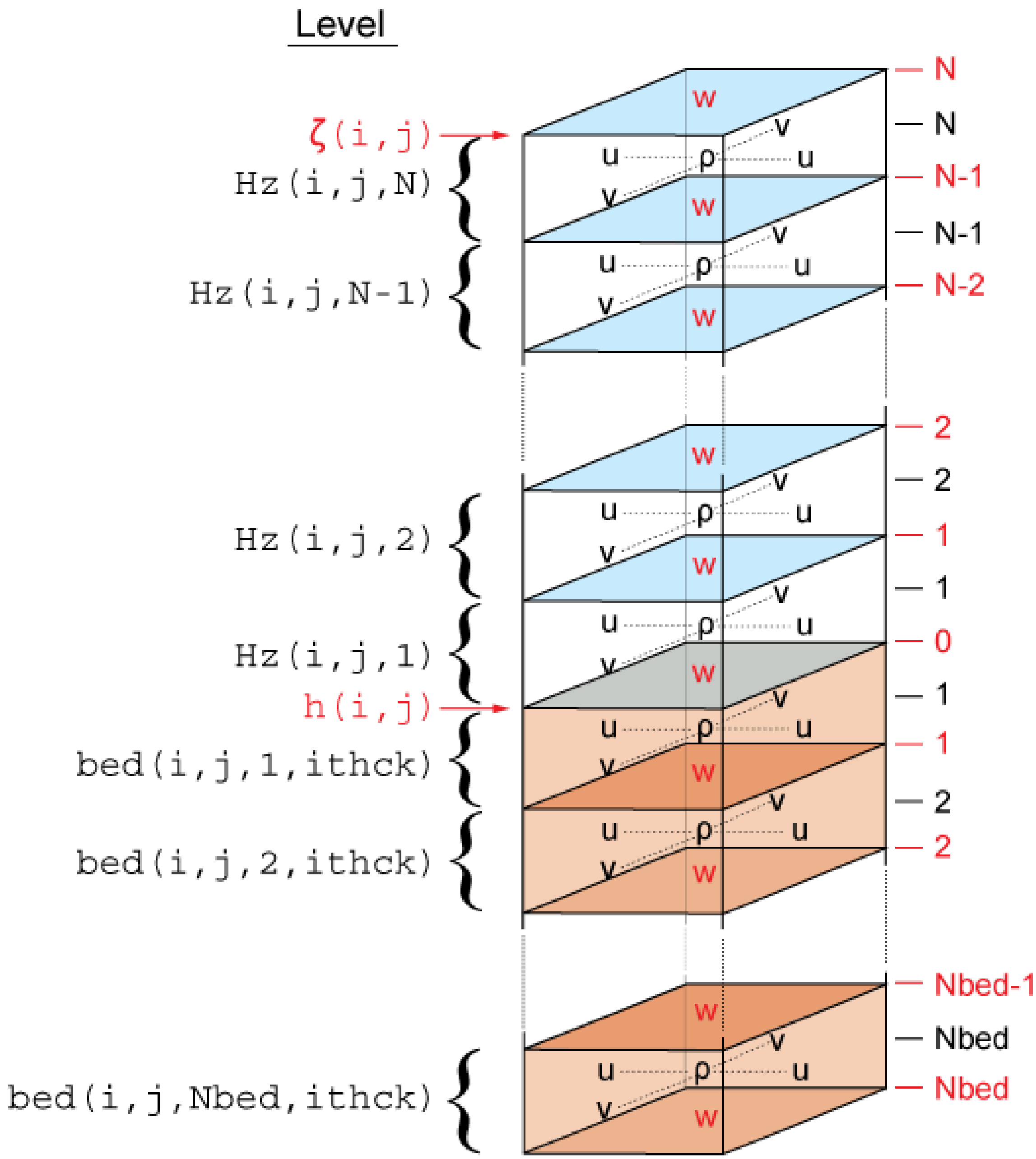


Figure 11.8: Vertical grid in the ROMS model

The vertical coordinate in ROMS is σ, which varies between -1 (ocean bottom) and 0 (ocean surface). There are two implementations of the σ to z conversion, both using sea-level anomalies (ζ) and bathymetry (h).

The first one is available in ROMS since 1999 and is given by:

$$z(x, y, \sigma, t) = S(x, y, \sigma) + \zeta(x, y, t)\left[1 + \frac{S(x, y, \sigma)}{h(x, y)}\right]$$

with

$$S(x, y, \sigma) = h_c\,\sigma + [h(x, y) - h_c]\; C(\sigma)$$

and h_c and $C(\sigma)$ parameters provided in the grid file.

The second transform, called UCLA-ROMS, is given by:

$$z(x, y, \sigma, t) = \zeta(x, y, t) + [\zeta(x, y, t) + h(x, y)]\; S(x, y, \sigma)$$

with

$$S(x, y, \sigma) = \frac{h_c\,\sigma + h(x, y)\,C(\sigma)}{h_c + h(x, y)}$$

and h_c and $C(\sigma)$ parameters provided in the grid file.

It can be rewritten in the same form as the original one.

$$z(x, y, \sigma, t) = h(x, y)S(x, y, \sigma) + \zeta(x, y, t) + \zeta(x, y, t)S(x, y, \sigma)$$

$$z(x, y, \sigma, t) = h(x, y)S(x, y, \sigma) + \zeta(x, y, t)\,[1 + S(x, y, \sigma)]$$

$$z(x, y, \sigma, t) = h(x, y)S(x, y, \sigma) + \zeta(x, y, t)\left[1 + \frac{h(x, y)S(x, y, \sigma)}{h(x, y)}\right]$$

In this form, both formulations can be expressed as:

$$z(x, y, \sigma, t) = H_0(x, y, \sigma) + \zeta(x, y, t)\left[1 + \frac{H_0(x, y, \sigma)}{h(x, y)}\right]$$

with H_0 which is constant overt time, and which varies between the classical and the UCLA formulations. For the classical formulation:

$$H_0(x, y, \sigma) = S(x, y, \sigma)$$

For the UCLA formulation:

$$H_0(x, y, \sigma) = h(x, y)S(x, y, \sigma)$$

11.0.3 MARS grid

In this section, the main features of the MARS grid and the implications in Ichthyop are summarized.

11.0.3.1 Horizontal

In MARS, the 3D structure of the grid is the same as the one in NEMO:

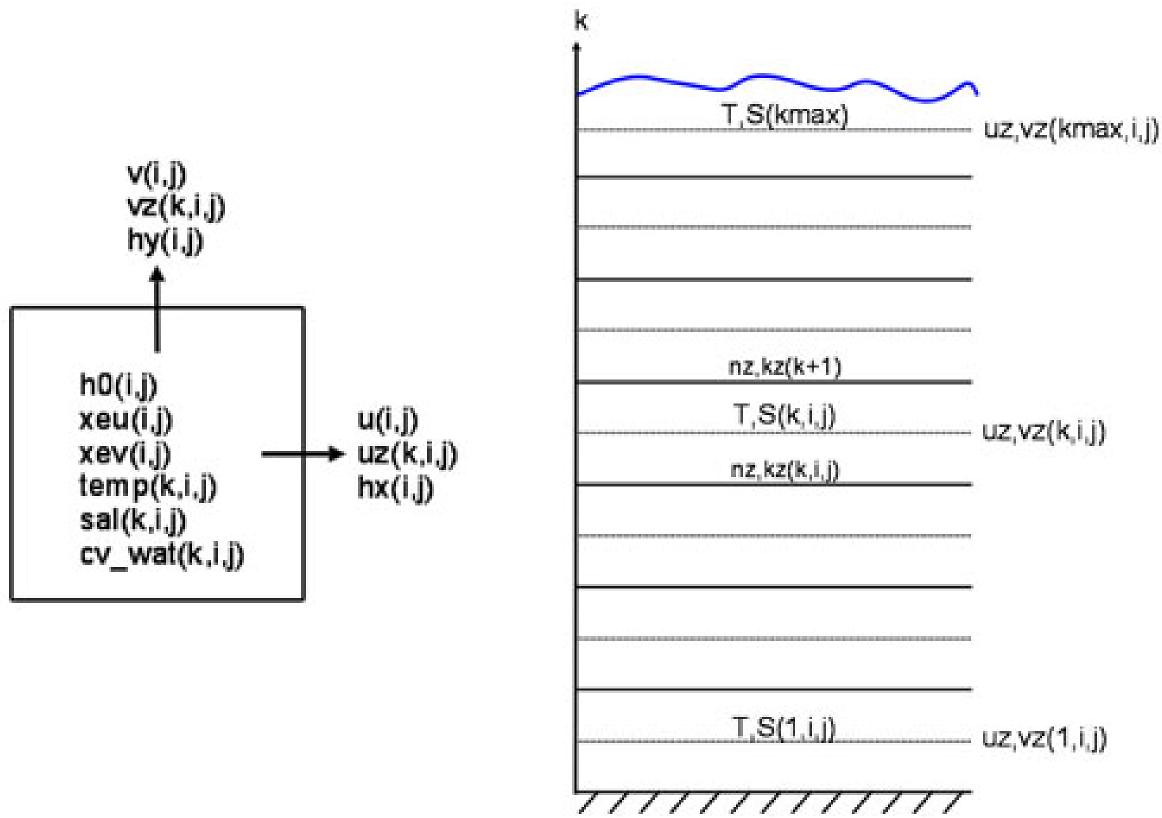


Figure 11.9: MARS staggered grid structure

11.0.3.2 Vertical

The vertical coordinate of the Mars model is called σ, which varies from `-1` at seabed to `0` at the surface. In MARS, if the number of `T` points on the vertical is `kmax`, the number of `W` points is `kmax + 1`. For a given `T` cell located at the `k` index, the corresponding `W` point is located below. `k=0` corresponds to the bottom, while `k=kmax` corresponds to the surface.

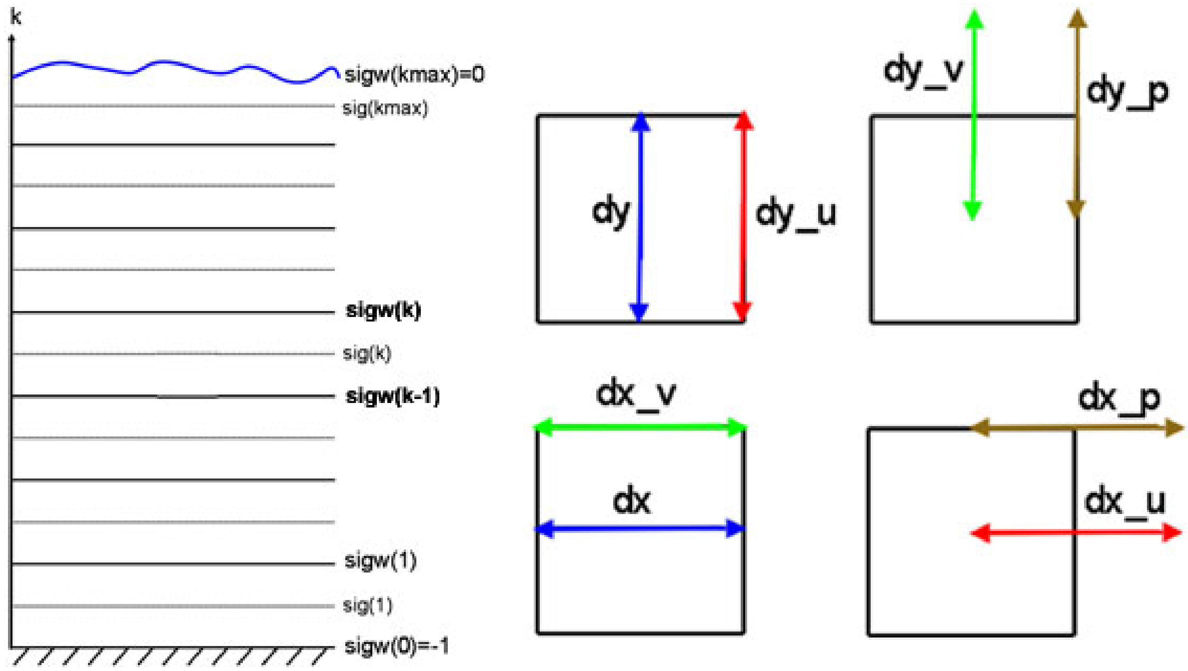


Figure 11.10: MARS grid structure.

The conversion from σ to z, using generalized σ levels, is given in Dumas (2009) (equation 1.29):

$$z = \xi(1+\sigma) + H_c \times [\sigma - C(\sigma)] + HC(\sigma)$$

where ξ is the free surface, H is the bottom depth and H_c is either the minimum depth or a shallow water depth above which we wish to have more resolution. C is defined as (equation 1.30):

$$C(\sigma) = (1-\beta)\frac{\sinh(\theta\sigma)}{\sinh(\theta)} + \beta\frac{\tanh[\theta(\sigma+\frac{1}{2})] - \tanh(\frac{\theta}{2})}{2\tanh(\frac{\theta}{2})}$$

However, if the H_c variable is not found, the following formulation will be used:

$$z = \xi(1 + \sigma) + \sigma H$$

Note that the $C(\sigma)$ variable is read from the input file.

12 Adding new processes

To Do

13 Adding output variable

When including new processes to Ichthyop (cf. Chapter 12), the storage of additional variable may be required. For instance, in the growth processes (see Section 7.3.8), in which particle length is a state variable, it is necessary to save length in the output NetCDF file. This is done by creating a new Java class associated with a property file.

13.1 Creating java class

Creating the Java class depends on what type of variable you want to save, as shown in Figure 13.1.

13.1.1 General case

Adding new variables can be achieved by creating new tracker class in the `org.previmer.ichthyop.io` package, which inherits from the `AbstractTracker` java class. It must override the 4 methods, as shown below for the `LengthTracker` class:

```java
public class LengthTracker extends AbstractTracker {

    @Override
    public void setDimensions() {
    }

    @Override
    public void addRuntimeAttributes() {
    }

    @Override
    public Array createArray() {
    }

    @Override
    public void track() {
    }
```

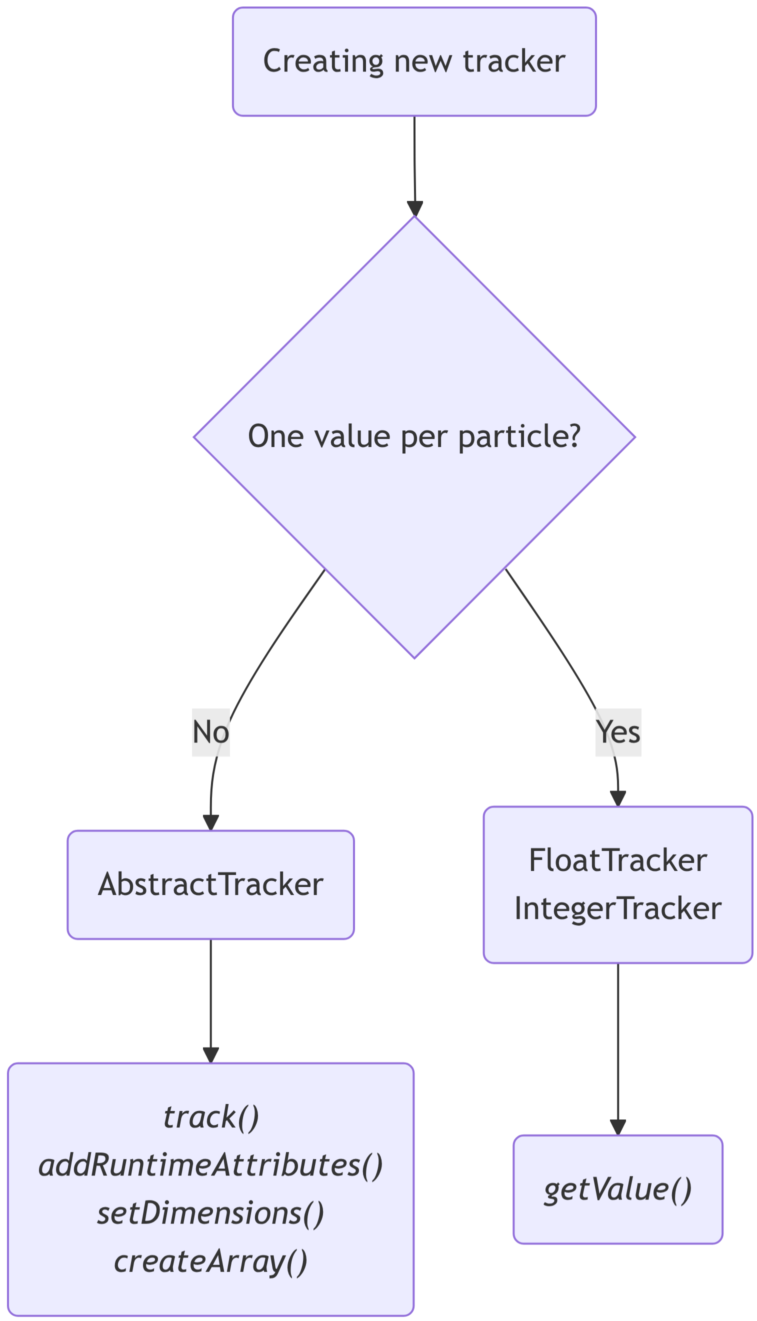


Figure 13.1: Adding a new tracker

```
}
```

`setDimensions` defines the dimensions associated with the variable. Time and drifter dimensions must be added by using the `addTimeDimension()` and `addDrifterDimension()` methods, respectively. A zone dimension can be added by calling the `addZoneDimension(TypeZone zoneType)` method. Custom dimensions can be added by using the `addCustomDimension(Dimension dim)` method.

`addRuntimeAttributes` defines additional attributes associated with the variable to be saved. Attributes are added by calling the `addAttribute(Attribute attribute)` method.

> **Note**
>
> Compulsory attributes and variable names are defined using properties files, see Section 13.2

`createArray` initializes the `Array` object that will be used to store the output variable. The dimensions of the array depends on the dimension of the output variables.

`track()` is the method that is called at each output time-step and which writes the variable in the NetCDF.

13.1.2 Simple case

Usually, new variables consist in tracking one single particle property, such as length for instance. In this case, the new tracker class can inherits either from the `FloatTracker` or the `IntegerTracker` classes as follows:

```
public class LengthTracker extends FloatTracker {

    @Override
    float getValue(IParticle particle) {
        ...
    }
}
```

In this case, the only method to define is the `getValue(IParticle particle)`, which specifies which particle's state variable is to be extracted for the given tracker.

13.2 Creating property file

In addition to the tracker java class, a property file must be included in the `io/resources/` folder. The name of this file must me the same as the Java class, except for the `.properties` suffix. For instance, the property file associated with the `LengthTracker.java` class will be named `LengthTracker.properties`. It must contain the following three lines:

```
tracker.shortname = length
tracker.longname = particle length
tracker.unit = millimeter
```

`tracker.shortname` is the name of the variable in the NetCDF, `tracker.longname` and `tracker.unit` are the values of the variable's longname and unit attributes.

14 References

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